Global Uncertainty, Local Credit: The Bank Lending Channel of International Spillovers[1]

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Abstract

Uncertainty about U.S. policy crosses borders through global banks. Matching supervisory data on euro area banks' U.S. exposures with the credit register, we exploit predetermined bank-level US exposure around the U.S. presidential elections and the time series variation of U.S. Economic and Policy uncertainty (EPU). Within the same firm and quarter, banks at the 75th versus 25th percentile of US asset exposure cut new lending at home by 9 percent after the 2024 presidential election, charge higher rates, shorten maturities, and reallocate their portfolio toward government EA securities. Effects are larger for banks with weak balance sheet characteristics and predict lower investment and employment at firms with indirect U.S. exposure via their banks.

Keywords: Economic policy uncertainty; bank lending channel; international spillovers; global banks; credit supply.
JEL classification: D80; E44; F34; F42; G21.

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1 Introduction

The transmission of global shocks across national borders is a central concern for both policymakers and academics. Given the dominant role of the United States in the international financial system, particular attention has been devoted to the international spillovers of U.S. macroeconomic and financial conditions. A growing literature documents how U.S. monetary policy and financial shocks propagate beyond domestic borders (see, e.g., Bruno and Shin, 2015; Miranda-Agrippino and Rey, 2020; Boehm and Kroner, 2026; Dedola, Rivolta and Stracca, 2017). Yet the role of uncertainty surrounding U.S. economic policy as a source of international spillovers has received comparatively less attention. This question has become particularly relevant given recent developments in U.S. fiscal, regulatory, and trade policies, which have generated considerable uncertainty regarding the future direction of economic policy. Understanding whether and how such uncertainty affects economic activity abroad is important for assessing the international transmission of shocks originating in the United States.

Economic policy uncertainty (EPU) is conceptually distinct from conventional monetary and financial shocks. Whereas changes in monetary policy or financial conditions directly affect borrowing costs, asset valuations, and financing constraints, uncertainty about future policies may influence economic decisions even before policy changes materialize. A central mechanism is the *wait-and-see* effect: when future returns become more uncertain and decisions are costly to reverse, economic agents have incentives to postpone investment and other long-term commitments until uncertainty is resolved (Bernanke, 1983; Rodrik, 1991; Bloom, 2009). A substantial literature documents the implications of uncertainty for corporate investment and domestic economic activity (see, e.g., Baker, Bloom and Davis, 2016; Gulen and Ion, 2016; Bloom, 2014). However, the channels through which economic policy uncertainty propagates internationally, particularly through financial intermediaries, remain less well understood.

Banks are natural candidates for transmitting these effects across borders. Globally active banks adjust their portfolios and lending decisions in response to foreign monetary, financial, and economic shocks (see, e.g., Cetorelli and Goldberg, 2011; Bruno and Shin, 2015; di Giovanni et al., 2022). Moreover, banks respond to heightened uncertainty by adjusting credit supply, lending standards, and portfolio composition (see, e.g., Buch, Buchholz and Tonzer, 2015; Gissler, Oldfather and Ruffino, 2016; Berger et al., 2022; Jasova, Mendicino and Supera, 2021). When uncertainty about U.S. economic policy increases, banks holding U.S. assets may face greater uncertainty about the future returns and risks of their foreign portfolios. In response, they may reduce their exposure to illiquid or risky assets, preserve balance-sheet capacity, and reallocate toward safer and more liquid investments. Importantly, these adjustments may extend beyond their U.S. operations and affect lending in their domestic markets. Financial integration may therefore transmit

uncertainty across borders through banks' foreign asset exposures, even to firms with limited direct economic links to the United States. Whether this channel operates in practice, and which banks and borrowers are most affected, are empirical questions that this paper addresses.

We investigate the international transmission of U.S. economic policy uncertainty to the euro area through banks' balance sheets and lending decisions. Specifically, we ask whether euro area banks with greater exposure to U.S. assets adjust their domestic credit supply more strongly following increases in U.S. policy uncertainty, and whether these adjustments affect the investment and employment decisions of their borrowers. We also investigate the mechanisms underlying banks' responses, distinguishing between precautionary adjustments in portfolio allocation and the amplification of credit contractions by pre-existing balance-sheet vulnerabilities.

Our analysis combines granular supervisory, credit-registry, and firm-level data. First, supervisory balance-sheet information for Significant Institutions under the Single Supervisory Mechanism (SSM) provides the geographical composition of banks' assets and allows us to measure their exposure to the United States. Second, the euro area credit registry (AnaCredit) provides detailed information on bank lending to non-financial corporations, including loan amounts, interest rates, maturities, and borrower risk. Crucially, the matched bank–firm structure allows us to compare the lending decisions of banks with different U.S. exposures to the same firm at the same point in time, thereby controlling for common borrower credit demand (Khwaja and Mian, 2008). Finally, we combine these data with firm balance sheets from Orbis and information on international trade linkages to examine the transmission of credit supply changes to firm-level outcomes and their interaction with borrowers' direct exposure to the United States.

Our empirical strategy proceeds in two complementary steps. First, we exploit episodes of heightened U.S. economic policy uncertainty associated with presidential elections, focusing particularly on the November 2024 election. We employ a difference-in-differences framework comparing lending outcomes before and after these events across banks with different predetermined exposures to U.S. assets. Our identifying assumption is that, absent the increase in U.S. policy uncertainty, lending by more and less exposed banks would have followed parallel trends, conditional on our controls and fixed effects. By comparing the credit supplied by different banks to the same firm within the same period, we isolate differential changes in bank credit supply from common changes in borrower demand. Second, we extend the analysis beyond individual events by constructing a time-varying measure of bank exposure to U.S. EPU, interacting banks' U.S. asset shares with the aggregate U.S. EPU index of Baker, Bloom and Davis (2016). This approach exploits both cross-sectional differences in banks' foreign exposures and time-series variation in U.S. policy uncertainty.

A central identification challenge is distinguishing the effects of policy uncertainty from changes in expected economic fundamentals. Presidential elections may simultaneously generate uncertainty and alter expectations about future trade, fiscal, and regulatory policies. Similarly, banks with greater U.S. exposures may respond differently to broader global financial shocks, independently of policy uncertainty. We address these concerns through several complementary exercises. We compare episodes characterized by different movements in U.S. EPU, including the 2016, 2020, and 2024 presidential elections and the escalation of U.S.–China trade tensions in 2018. We also examine whether our results are robust to controlling for U.S. macroeconomic conditions, financial market volatility, monetary policy uncertainty, geopolitical risk, and other sources of global shocks. Finally, we distinguish banks' exposure through U.S. assets from their exposure through U.S. liabilities, allowing us to investigate whether the transmission operates primarily through foreign asset holdings rather than funding relationships. Together, these tests help assess whether the estimated lending responses reflect exposure to U.S. policy uncertainty rather than broader changes in international economic and financial conditions.

Our results provide evidence that increases in U.S. economic policy uncertainty are associated with a contraction in domestic credit supply by euro area banks with greater U.S. asset exposures. Following the November 2024 election, banks with higher predetermined U.S. exposures reduce lending to euro area non-financial corporations relative to less exposed banks. Within the same firm and quarter, moving from the 25th to the 75th percentile of bank U.S. exposure is associated with approximately 8.9% lower new lending. The contraction is accompanied by higher lending rates, shorter loan maturities, and a greater probability of terminating existing bank–firm relationships. These differences emerge following the election, with no evidence of differential pre-trends, and are already visible before the introduction of major U.S. tariff measures in April 2025.

The contraction in corporate lending is accompanied by changes in banks' portfolio composition. More exposed banks reduce their aggregate lending to euro area firms while increasing their holdings of government securities. This pattern suggests that the response is not simply a reallocation of credit across domestic borrowers, but reflects a broader adjustment in banks' asset allocation. Consistent with a precautionary response to heightened uncertainty, banks reduce exposure to relatively illiquid lending commitments while increasing their holdings of more liquid securities. We also find that U.S. asset exposure, rather than U.S. liability exposure, is the relevant dimension for explaining differential credit supply responses, supporting the role of foreign asset holdings in the international transmission of uncertainty.

The analysis based on continuous variation in U.S. EPU confirms and extends these findings. A one-standard-deviation increase in bank-level exposure to U.S. policy uncertainty is associated with an approximately 4.5% reduction in new lending to euro area

non-financial firms. In addition, annualized lending rates increase by approximately 90 basis points, loan maturities decline by around 70 days, and the probability of terminating a bank–firm relationship increases by 0.8 percentage points. These results remain robust to demanding specifications incorporating firm×time and bank–firm fixed effects, as well as controls for alternative sources of international economic and financial shocks. Taken together, the event-study and panel estimates indicate that banks' exposure to U.S. economic policy uncertainty affects not only the quantity of credit supplied but also its pricing, maturity, and continuity.

We next examine the mechanisms underlying these adjustments. Our results point to two complementary channels. First, the reduction in lending commitments, shortening of loan maturities, and shift toward more liquid assets are consistent with a *wait-and-see* channel, whereby banks respond to heightened uncertainty by limiting their exposure to assets that are relatively costly to adjust. Second, pre-existing bank vulnerabilities amplify the credit contraction. Banks with higher non-performing loan ratios and lower liquidity buffers exhibit stronger lending responses, consistent with tighter balance-sheet constraints increasing the cost of bearing additional risk. By contrast, we find no statistically significant moderating role for regulatory capital ratios. On the borrower side, credit contractions are more pronounced for riskier firms and for firms with greater exposure to U.S. markets. These patterns suggest that both bank balance-sheet conditions and borrower characteristics shape the distribution of international uncertainty spillovers.

Finally, we investigate whether these changes in bank lending are associated with real economic outcomes. We construct a firm-level measure of exposure to U.S. EPU based on the exposure of each firm's lenders, weighted by their shares in the firm's total borrowing. Firms moving from the 25th to the 75th percentile of this exposure experience approximately 7.9% lower new bank borrowing, 2% lower investment, and 0.5% lower employment, with no evidence of a compensating increase in other debt. These effects are stronger among firms with greater direct trade exposure to the United States, suggesting that financial and trade linkages can reinforce one another. Importantly, the negative responses are not confined to firms trading directly with the United States, indicating that bank lending provides an additional channel through which U.S. policy uncertainty may affect euro area economic activity. While firm-level estimates do not exploit the same within-firm identification as our loan-level analysis, their consistency with the credit supply evidence supports the interpretation that internationally exposed banks contribute to the propagation of foreign uncertainty to domestic real activity.

Related literature and contribution. Our paper contributes to three strands of the literature. First, we contribute to the literature on the international transmission of financial shocks through global banks. Existing studies document how shocks to bank

funding, capital, monetary conditions, and foreign borrowers propagate across borders and affect domestic credit supply (Peek and Rosengren, 1997, 2000; Cetorelli and Goldberg, 2011; Schnabl, 2012; Ongena, Peydró and van Horen, 2015; Popov and Udell, 2012; Bruno and Shin, 2015; di Giovanni et al., 2022). We extend this literature by examining uncertainty about foreign economic policies as a potential source of international financial spillovers. In contrast to studies focusing primarily on realized financial shocks or changes in borrower fundamentals, we investigate how changes in uncertainty surrounding foreign policy developments affect the lending decisions of banks exposed through their asset holdings. Our findings also speak to the literature on banks' international portfolio allocation (Giannetti and Laeven, 2012), showing that heightened uncertainty about foreign assets can be associated with a reallocation toward liquid securities rather than increased lending to domestic borrowers.

Second, we contribute to the literature on uncertainty and financial intermediation. A growing body of evidence shows that economic policy uncertainty affects corporate investment, bank risk-taking, and domestic credit supply (Bernanke, 1983; Bloom, 2009; Baker, Bloom and Davis, 2016; Buch, Buchholz and Tonzer, 2015; Gissler, Oldfather and Ruffino, 2016; Jasova, Mendicino and Supera, 2021; Berger et al., 2022). We extend this work to an international setting by showing that uncertainty about U.S. economic policy is associated with changes in the lending behavior of euro area banks holding U.S. assets. The richness of our data allows us to characterize the response along several margins, including lending volumes, interest rates, maturities, relationship termination, and securities holdings. We further document how banks' pre-existing vulnerabilities and borrower risk influence the magnitude of these responses.

Third, our paper relates to the literature identifying bank credit supply shocks using matched bank–firm data and tracing their effects to the real economy (Khwaja and Mian, 2008; Jiménez et al., 2012; Chodorow-Reich, 2014; Altavilla, Boucinha and Peydró, 2018; Altavilla, Canova and Ciccarelli, 2020). We apply this identification strategy to the international transmission of economic policy uncertainty, exploiting heterogeneity in banks' foreign asset exposures while controlling for borrower demand. By combining loan-level evidence with firm-level investment and employment outcomes, we show how financial integration may transmit uncertainty originating abroad to domestic firms, including those without direct trade relationships with the source country.

The remainder of the paper is organized as follows. Section 2 discusses the economic mechanisms. Section 3 describes the data and measures of exposure to U.S. economic policy uncertainty. Section 4 presents the event-study analysis. Section 5 examines the effects of continuous variation in bank-level uncertainty exposure and the underlying transmission mechanisms. Section 6 investigates firm-level outcomes. Section 7 presents tests distinguishing policy uncertainty from alternative sources of international shocks.

Section 8 discusses robustness checks, and Section 9 concludes.

2 Conceptual framework and hypotheses

This section lays out, informally, the mechanism through which uncertainty about U.S. economic policy can be transmitted to credit conditions in the euro area by banks that hold U.S. assets, and it derives the hypotheses that guide the empirical analysis.

From U.S. policy uncertainty to bank balance sheets. Consider a euro area bank that holds a portfolio of U.S. assets—loans to U.S. borrowers, U.S. securities, derivatives with U.S. counterparties—alongside its domestic loan book. An increase in uncertainty about U.S. fiscal, regulatory, or trade policy raises the dispersion of the future returns on these assets: the creditworthiness of U.S. borrowers, the valuation of U.S. securities, and the regulatory treatment of U.S. exposures all become harder to predict, even if their expected values are unchanged. The bank's perceived portfolio risk therefore rises in proportion to its U.S. exposure. It is useful to distinguish this second-moment effect from *bad news*, which lowers the expected value of the same assets. Both raise the risk that the bank's internal models assign to its U.S. borrowers, and both can lead the bank to retrench; but only uncertainty raises the value of waiting and therefore predicts a tilt toward reversible positions rather than a uniform reduction in risk-taking, and only bad news predicts a response that carries the sign of the news. Under standard risk-management practices, a rise in the riskiness of one part of the portfolio reduces the capacity of the bank to take on risk elsewhere, including at home.

Wait-and-see. The real-options literature emphasizes that when returns are uncertain and commitments are costly to reverse, the value of waiting rises (Bernanke, 1983; Rodrik, 1991; Bloom, 2009). For a lender, a new term loan to a firm is a largely irreversible commitment: it ties up balance-sheet capacity for years, cannot be resold without a discount, and its credit risk is only revealed slowly. When uncertainty about the bank's own portfolio rises, the bank has an incentive to preserve optionality by postponing such commitments, by granting loans of shorter maturity, by pricing in a higher risk premium, and by shifting toward assets that are liquid and easily reversible, such as government securities. Under this *wait-and-see* channel, the bank tightens credit supply to domestic firms not because their creditworthiness has changed, but because the bank's own willingness to lock in capacity has fallen.

Balance-sheet amplification. The strength of the response should depend on the bank's initial condition. Models of the bank lending channel predict that banks with

weaker balance sheets—lower asset quality, thinner liquidity buffers, or binding regulatory constraints—face a higher marginal cost of intermediation and respond more strongly to adverse shocks (Bernanke and Gertler, 1995; Kashyap and Stein, 2000). Uncertainty tightens internal risk limits and value-at-risk constraints, which bind sooner for banks whose portfolios are already impaired. Theories of precautionary liquidity hoarding under uncertainty similarly imply that banks with limited liquidity buffers cut lending more aggressively (Berger et al., 2022). Under this *balance-sheet amplification* channel, the credit contraction induced by a given increase in U.S. uncertainty is larger for vulnerable banks.

Borrower heterogeneity. A bank that reduces its risk-taking capacity should rebalance away from the borrowers whose repayment prospects are most sensitive to the source of the uncertainty. Two groups stand out: borrowers with high ex ante default probabilities, whose loans absorb the most risk capacity, and borrowers whose cash flows are directly exposed to the United States through trade, for whom U.S. policy uncertainty and the bank's own portfolio uncertainty are correlated. If the bank lending channel of foreign uncertainty operates as described, these borrowers should experience the largest credit contractions.

Real effects. Finally, if the credit contraction is not offset by substitution toward other lenders or funding sources, firms that borrow from exposed banks should reduce investment and employment. Because bank relationships are sticky and unlisted firms rely predominantly on bank finance, substitution is likely to be incomplete, particularly for smaller firms.

Discriminating predictions. The two channels are complementary, but they have different signatures, and we organize the mechanism evidence accordingly. The wait-and-see channel predicts (i) shorter maturities on the loans that are granted, (ii) a larger response of irreversible commitments, such as term loans, than of adjustable ones, such as credit lines, and (iii) a reallocation toward liquid assets that occurs before, or without, realized losses. The balance-sheet amplification channel predicts (iv) a larger response where risk constraints bind, at banks with impaired asset quality or thin liquidity, and (v) a retreat from the borrowers that absorb the most risk capacity. Evidence of type (iv)–(v) is consistent with ordinary credit-risk tightening and does not, by itself, establish wait-and-see behavior; evidence of type (i)–(iii), together with the absence of a response when expected policy changes but uncertainty does not, is what distinguishes a response to uncertainty from a response to a first-moment deterioration in asset quality.

Testable hypotheses. The framework yields four hypotheses that structure the analysis:

H1. *First stage.* An increase in U.S. policy uncertainty raises the risk that banks with larger U.S. exposure assign to their U.S. portfolios, as measured by the probabilities of default in their internal models.

H2. *Credit supply.* Conditional on borrower demand, banks with larger U.S. exposure reduce new lending to euro area firms, charge higher rates, shorten maturities, and terminate more relationships when U.S. policy uncertainty rises; they reallocate their balance sheets toward safer and more liquid assets. Reversible instruments, such as credit lines, respond less.

H3. *Amplification and reallocation.* The contraction is larger for banks with weaker asset quality and thinner liquidity buffers, and it is directed disproportionately at riskier borrowers and at borrowers with stronger trade links to the United States.

H4. *Real effects.* Firms whose lenders are more exposed to U.S. policy uncertainty reduce borrowing, investment, and employment.

Two features of the framework guide the identification strategy. First, the shock enters through the *asset* side of banks' balance sheets: a bank's exposure to U.S. policy uncertainty should be measured by its U.S. assets, not by its U.S. funding. We test this directly by contrasting asset- and liability-based exposure measures. Second, the mechanism should be triggered by increases in *uncertainty*, whatever their political origin. We therefore compare election episodes that did and did not raise U.S. EPU, and an uncertainty episode that did not coincide with an election. Third, and relatedly, the mechanism responds to the *second* moment of U.S. policy, not to its expected direction: an episode that changes expected U.S. policy without raising uncertainty about it should leave exposed banks' lending unchanged, and the response to an uncertainty shock should not carry the sign of the accompanying news about U.S. asset values. Section 4.6 organizes the event-study evidence around these implications.

3 Data and measurement

We combine several confidential and public data sources into a unique dataset that allows us to follow the transmission of U.S. policy uncertainty from the balance sheets of euro area banks to their lending decisions and, ultimately, to the investment and employment decisions of their borrowers. Our sample covers 79 euro area banking groups classified as Significant Institutions under the Single Supervisory Mechanism (SSM), comprising roughly 500 individual credit institutions, over the period 2014Q4–2025Q2. Table 1 reports summary statistics for the main variables.

3.1 Data sources

Loan-level data. Our primary source of credit data is AnaCredit, the euro area credit register maintained by the European System of Central Banks. AnaCredit records, at monthly frequency, all credit exposures of euro area credit institutions to legal entities, with a reporting threshold of €25,000 on the total exposure of a debtor to a reporting institution.[2] For each instrument, the register reports the outstanding amount, the type of instrument (term loans, credit lines, revolving credit, and others), the date of inception, the original and residual maturity, the annualized agreed interest rate, the currency of denomination, and the probability of default (PD) assigned to the debtor by the lending bank under its internal risk models. It also reports borrower characteristics such as location, sector of activity (NACE), and size. We collapse the data to the bank–firm–quarter level and focus on *new* credit, defined as the volume of instruments whose inception date falls in the quarter, since new lending is the margin along which credit supply adjusts most quickly. Loan rates and maturities are aggregated to the bank–firm–quarter level as loan-size-weighted averages of the instruments originated in the quarter. This level of granularity allows us to trace how banks adjust corporate lending along several margins of the loan contract in response to higher foreign policy uncertainty.[3]

Supervisory data. We match the credit register with bank-level supervisory information from the Common Reporting (COREP) and Financial Reporting (FINREP) frameworks maintained by the SSM. FINREP templates F20.04 and F20.06 provide, at the level of the consolidated banking group, the geographical breakdown of assets and liabilities by residence of the counterparty. These templates are the source of our measure of banks' exposure to the United States. We define a group's *U.S. exposure* as the sum of its derivatives, equity instruments, cash balances at central banks and other demand deposits, debt securities, and loans and advances vis-à-vis counterparties resident in the United States, scaled by total assets. FINREP also provides the balance-sheet items we use as time-varying controls and as outcomes in the bank-level analysis: total assets, loans to non-financial corporations by counterparty region, holdings of government securities by

[2] Because the threshold applies to the debtor's total exposure to a bank, individual loans below €25,000 appear in the data whenever the debtor's overall exposure exceeds the threshold.

[3] AnaCredit reporting began in September 2018, but each instrument carries its date of inception, so originations from 2014Q4 onward are observed for all instruments that were still outstanding when reporting started. The analyses of the 2016 election and of the 2018 trade war in Section 4.6 rely on these pre-2018 originations. Because instruments that had been fully repaid before September 2018 are not observed, the pre-2018 sample of new loans under-represents short-maturity lending. This selection is not innocuous for our purposes: if exposed banks respond to uncertainty by shortening maturities, their pre-2018 originations are less likely to survive into the reporting window, which would overstate the estimated contraction in their new lending and bias the maturity estimate toward zero. We therefore treat the pre-2018 loan-level estimates as corroborating evidence, rely on contemporaneous supervisory (FINREP) bank-level outcomes for the 2016 episode, which are not subject to this selection, and let the 2024 event and the post-2018 panel carry the causal burden.

issuer region, non-performing loans, cash, and return on assets. COREP reports prudential metrics, including the Common Equity Tier 1 (CET1) ratio and the Liquidity Coverage Ratio (LCR).

Firm data. We obtain financial information on corporate borrowers from Bureau van Dijk's Orbis database, which has good coverage of unlisted firms and small and medium-sized enterprises in the euro area. We use total assets, cash holdings, tangible assets, sales, and the interest coverage ratio as firm-level controls, and fixed assets, employment, bank loans, and total debt as outcomes in the analysis of real effects.

Group structure. We use the Register of Institutions and Affiliates Database (RIAD) maintained by the ECB to map individual credit institutions to their consolidated banking groups and to identify the ownership structure of financial and non-financial corporations in the sample.

Trade exposure. We use annual data from the OECD Inter-Country Input-Output (ICIO) tables to measure, at the country–sector level, the gross exports to and imports from the United States of the sectors in which our borrowers operate. For a country c and NACE two-digit sector s in year t, we define net exports to the United States as

$$\text{Net Exports to US}_{c,s,t} = \frac{\text{EX}_{c,s\to US,t} - \text{IM}_{c,s\leftarrow US,t}}{\sum_{c'} \left(\text{EX}_{c,s\to c',t} + \text{IM}_{c,s\leftarrow c',t}\right)}, \tag{1}$$

that is, the sector's net exports to the United States relative to its total trade. This variable characterizes borrowers' own exposure to the U.S. economy.

Uncertainty indices. Our measure of U.S. economic policy uncertainty is the monthly index of Baker, Bloom and Davis (2016), which tracks the frequency of articles in ten leading U.S. newspapers that contain terms related to "uncertainty," "economy," and "policy." In robustness checks and in the analysis of alternative sources of risk, we also use the U.S.–China trade tension index of Rogers, Sun and Sun (2024), the U.S. monetary policy uncertainty index of Husted, Rogers and Sun (2020), the geopolitical risk index of Caldara and Iacoviello (2022), the country-level EPU indices of Baker, Bloom and Davis (2016) for euro area countries and for major non-euro-area economies, and the VIX.

3.2 Euro area banks' exposure to the United States

Figure 2 documents the geographical composition of euro area banks' extra-euro-area exposures at the end of 2024. Exposures to countries outside the euro area amount on average to about 21 percent of total assets. The United States is by far the largest single

foreign counterparty country: U.S. exposures average 4.3 percent of total assets, ahead of the United Kingdom (3.5 percent) and well ahead of any other country. The United States thus accounts for roughly one fifth of euro area banks' international exposures.

There is sizable heterogeneity in U.S. exposure across banks, which is the source of variation we exploit. Figure 3 shows that average U.S. exposure ranges from below 3 percent of total assets for groups headquartered in Italy, Austria, and Finland to close to 9 percent for groups headquartered in Germany and Ireland (panel (a)); that the cross-sectional distribution is right-skewed, with a large mass of banks with little or no U.S. exposure and a tail of banks with exposures above 15 percent of total assets (panel (b)); and that exposure increases with bank size, from 1 percent of total assets in the bottom quintile to almost 8 percent in the top quintile (panel (c)). Over the sample period, average U.S. exposure declined from about 4.5 percent of total assets in 2015 to below 3 percent in 2020 and recovered to about 4 percent by 2024 (Appendix Figure A2). Importantly for identification, the time series of the U.S. EPU index and of the average U.S. exposure of euro area banks are negatively correlated (correlation coefficient -0.37): banks did not, on average, increase their U.S. exposure in anticipation of periods of high uncertainty.

3.3 Bank-level exposure to U.S. policy uncertainty

For the panel analysis of Section 5, we combine bank-level U.S. exposure with the time series of U.S. EPU into a single bank–quarter measure:

$$BEPU_{b,t-1} = \frac{1}{4}\left(\sum_{i=1}^{4} \frac{\text{US Exposure}_{b,t-i}}{\text{Total Assets}_{b,t-i}}\right) \times \text{US EPU}_{t-1}. \tag{2}$$

The first component is a smoothed measure of bank b's exposure to the U.S. economy, computed as the average share of U.S. assets in total assets over the previous four quarters. Averaging over time mitigates concerns about short-term balance-sheet adjustments and measurement error, and ensures that exposure is predetermined with respect to lending decisions in quarter t. The second component is the level of the U.S. EPU index in the previous quarter (in the baseline, the quarterly average of the monthly index; Section 8 considers alternative timing conventions). $BEPU_{b,t-1}$ can be interpreted as a shift-share variable in which a common aggregate shock, U.S. policy uncertainty, is loaded onto banks according to their predetermined balance-sheet exposure.

BEPU has a mean of 5.8 and a standard deviation of 7.4; moving from the 25th to the 75th percentile of its distribution corresponds to an increase of about 7.3, roughly one standard deviation, which is the metric we use to quantify effects throughout. A variance decomposition (Appendix Table A6) shows that about 11 percent of the total variation in

BEPU is common across banks within a quarter, while 89 percent reflects cross-sectional differences across banks within the same quarter. This cross-sectional component is what survives the firm×quarter fixed effects used in our regressions. Appendix Table A7 shows that, conditional on bank and time fixed effects, *BEPU* is higher for larger, better-capitalized, and globally systemically important banks, and lower for banks with higher liquidity coverage and profitability. These correlations motivate the inclusion of time-varying bank controls in all specifications.

4 Event-study evidence: U.S. elections as uncertainty shocks

We begin with a difference-in-differences design that exploits sharp and plausibly exogenous increases in U.S. policy uncertainty around U.S. presidential elections. The advantage of this design is transparency: the timing of the shock is known, exposure is measured before the shock, and pre-trends can be inspected directly. Our main event is the November 2024 election, which produced the largest surge in U.S. EPU in the history of the index (Figure 1). Section 4.6 repeats the analysis around the 2016 election, the 2020 election, and the 2018 escalation of the U.S.–China trade war.

4.1 Empirical design

The key source of identification is cross-sectional heterogeneity in euro area banks' pre-determined exposure to the United States. We compare lending outcomes of banks with higher and lower U.S. exposure before and after the election-induced increase in uncertainty. Our baseline specification at the bank–firm–quarter level is

$$\log(\text{Credit})_{b,f,t} = \beta\, \text{Post}_t \times \text{US Exp}_b + \Gamma X_{b,t-1} + \alpha_{f,t} + \alpha_{b,f} + \epsilon_{b,f,t}, \tag{3}$$

where $\log(\text{Credit})_{b,f,t}$ is the logarithm of new credit granted by bank b to firm f in quarter t, Post_t is an indicator equal to one in the post-election quarters, and US Exp_b is bank b's U.S. exposure, measured as U.S. assets over total assets in 2023Q3—one year before the election and before the start of the campaign—so that it is not affected by the shock itself. For the 2024 election, the pre-period is 2023Q4–2024Q3 and the post-period is 2024Q4–2025Q2. The post-period includes 2025Q2, the quarter of the April 2025 tariff announcements, which constitute a second large uncertainty shock; the dynamic estimates below show that the contraction is already fully present in 2024Q4 and 2025Q1, so the election effect does not depend on the last quarter of the window.

The coefficient of interest, β, captures the differential change in lending after the

election for banks with greater pre-election U.S. exposure. To interpret it as a credit supply effect, we saturate the model with fixed effects. Firm×quarter fixed effects $\alpha_{f,t}$ absorb all time-varying firm-level determinants of credit demand as well as aggregate and sectoral conditions (Khwaja and Mian, 2008): identification comes from comparing, within the same firm and quarter, the lending of banks with different U.S. exposure. This requires firms with multiple lending relationships, which are prevalent among euro area corporate borrowers. Bank×firm fixed effects $\alpha_{b,f}$ absorb all time-invariant characteristics of the lending relationship, such as screening, matching, and long-term contracting. The vector $X_{b,t-1}$ contains time-varying bank controls lagged one quarter to mitigate simultaneity concerns: the logarithm of total assets, the CET1 ratio, the liquidity coverage ratio, return on assets, and the non-performing loan ratio. Standard errors are clustered at the bank–quarter level to allow for correlated shocks across a bank's lending relationships within a quarter. The identifying assumption is that, absent the increase in U.S. EPU, the lending of more and less exposed banks to the same firm would have followed parallel trends. We assess this assumption below by estimating a dynamic version of (3).

4.2 First stage: exposed banks experience higher credit risk in the United States

Hypothesis H1 states that U.S. policy uncertainty raises the riskiness of exposed banks' U.S. portfolios. We test it using the PDs that banks assign to their borrowers, aggregated to the bank–country–quarter level as exposure-weighted averages, in the specification

$$\log(\text{PD})_{b,c,t} = \beta\, \text{Post}_t \times \text{High US Exp}_b + \Gamma X_{b,t-1} + \alpha_t + \alpha_b + \alpha_c + \epsilon_{b,c,t}, \tag{4}$$

where High US Exp_b identifies banks whose 2023Q3 U.S. exposure lies above the 75th percentile of the pre-treatment distribution, and α_t, α_b, and α_c are quarter, bank, and borrower-country fixed effects.

Table 2 reports the results. Column (1) pools all borrower countries: after the 2024 election, the average borrower PD of highly exposed banks increases by about 0.46 log points relative to other banks, significant at the 10 percent level. Columns (2)–(4) split the portfolio by borrower region. The increase is concentrated in the U.S. loan book, where borrower PDs of highly exposed banks rise by about 0.59 log points (column 2, significant at the 1 percent level), while the differential changes in the euro area and rest-of-world portfolios are smaller and not statistically significant. The shock therefore enters exposed banks' balance sheets exactly where the framework predicts: through the perceived riskiness of their U.S. assets. One caveat applies. PDs are the expected default frequencies assigned by banks' internal models, so they measure perceived credit risk rather than the dispersion of outcomes; the first stage shows that exposed banks perceive

their U.S. portfolios as riskier, not, by itself, that uncertainty rather than expected losses drives that perception. The evidence that separates the two is presented in Section 4.6. Appendix Table A5, which uses the continuous exposure measure in a bank-level regression, confirms the pattern.

4.3 Higher U.S. EPU reduces bank lending to euro area firms

Table 3 presents the estimates of equation (3). Columns (1)–(3) consider lending volumes. The coefficient on Post $\times$ US Exp_b is negative and statistically significant at the 1 percent level for total new credit and for term loans: more exposed banks internalized the increase in U.S. policy uncertainty and reduced their new lending to the same euro area firms by more than less exposed banks did. The magnitudes are economically meaningful. Moving from the 25th to the 75th percentile of the pre-election distribution of U.S. exposure is associated with a reduction in new total credit of about 8.9 percent and in new term loans of about 8.7 percent in the three quarters following the election. The coefficient for credit lines (column 3) is small, positive, and imprecisely estimated: its standard error is almost five times that of the term-loan coefficient, and its confidence interval includes the term-loan estimate, so we cannot reject that credit lines respond as much as term loans. The point estimates are nonetheless in line with the wait-and-see channel, under which term loans, which lock in balance-sheet capacity for years, should respond more than credit lines, which are shorter, renewable, and drawn at the borrower's discretion within pre-committed limits; the panel evidence in Section 5.2 sharpens this comparison.

Columns (4)–(6) turn to non-price and price terms of lending. Exposed banks are more likely to terminate existing relationships: the coefficient on the exit indicator, which equals one when a bank–firm relationship active in $t-1$ ceases to exist in t, is positive and significant, implying that a 25th-to-75th percentile increase in exposure raises the probability of termination by about 1.8 percentage points. Exposed banks also shorten the maturity of the loans they do grant, by about 2.4 percent (roughly 35 days), and charge higher annualized loan rates, by about 3.4 basis points. Higher U.S. EPU triggered by the 2024 election thus tightened credit conditions for euro area firms along every margin of the loan contract: quantity, price, maturity, and the extensive margin of relationships.

Dynamics and parallel trends. To trace the dynamics of the response and assess the validity of the parallel-trends assumption, we estimate an event-study version of (3) that replaces Post_t with a full set of quarter indicators:

$$\log(\text{Credit})_{b,f,t} = \sum_{k \neq 2024Q3} \beta_k \, \text{US Exp}_b \times \mathbb{1}\{t = k\} + \Gamma X_{b,t-1} + \alpha_{f,t} + \alpha_b + \epsilon_{b,f,t}, \quad (5)$$

where the quarter immediately preceding the election, 2024Q3, is the omitted category. Figure 4 plots the estimated β_k with 90 percent confidence intervals. In the pre-election quarters, the coefficients are small and not statistically distinguishable from zero: there is no differential trend in lending between more and less exposed banks before the shock. The contraction materializes in 2024Q4, the quarter of the election, when the coefficient drops sharply, and it persists through 2025Q2, with coefficients that remain negative and significant at conventional levels. Because the contraction is already fully present in 2024Q4 and 2025Q1, it cannot be attributed to the April 2025 tariff announcements, which fall in the last quarter of the window. The absence of pre-trends and the timing of the response support a causal interpretation.

Asset-side versus liability-side exposure. The framework implies that the relevant exposure is on the asset side of the balance sheet. This distinguishes our channel from the well-documented transmission of foreign *funding* shocks, which operates through banks' liabilities and is especially relevant in crisis times (Schnabl, 2012; Carpinelli and Crosignani, 2021). Table 4 contrasts the two. Column (1) repeats the baseline asset-based estimate. Column (2) adds the lagged, time-varying share of U.S. liabilities in total assets as a control: the coefficient on Post $\times$ US Assets/TA$_b$ remains negative and statistically significant, while U.S. liabilities enter with a negative level effect. Column (3) replaces the treatment variable with the pre-election share of U.S. liabilities in total assets: the interaction is small, positive, and not significantly different from zero. The transmission of U.S. policy uncertainty to euro area lending therefore runs through banks' U.S. assets rather than through their U.S. funding, consistent with euro area banks holding ample, if declining, liquidity buffers over the sample period.

4.4 Heterogeneity across banks and firms

Hypothesis H3 predicts that the contraction is amplified for vulnerable banks and directed at the borrowers most sensitive to U.S. uncertainty. We augment (3) with a triple interaction between Post$_t$ $\times$ US Exp$_b$ and an indicator for a bank or firm characteristic. Table 5 reports the results. On the bank side (columns 1–2), the contraction is significantly stronger for banks with non-performing loan ratios above the 75th percentile: the triple interaction is negative and significant at the 5 percent level, and its magnitude is large relative to the baseline effect. By contrast, the triple interaction for banks with capital ratios below the 25th percentile is negative but imprecisely estimated, while the baseline effect remains significant; given its standard error, we cannot reject that it equals the NPL interaction. The evidence therefore identifies asset quality as an amplifying margin without establishing that regulatory capital is not one. On the firm side (columns 3–4), exposed banks cut credit by more to riskier firms, identified by a PD above the 75th

percentile, and to firms operating in sectors whose export exposure to the United States lies above the 95th percentile. Both triple interactions are negative and significant at the 10 percent level. Banks thus reallocate credit away from borrowers whose repayment capacity is most sensitive to U.S. policy developments.

4.5 Bank-level portfolio reallocation

The loan-level estimates identify a relative contraction in credit supply across banks lending to the same firm. They do not, by themselves, establish that exposed banks reduced their overall lending, since a bank could in principle reallocate credit from multi-bank firms to other borrowers. We therefore estimate the bank-level counterpart of (3),

$$Y_{b,t} = \beta \, \text{Post}_t \times \text{US Exp}_b + \Gamma X_{b,t-1} + \alpha_t + \alpha_b + \epsilon_{b,t}, \tag{6}$$

where $Y_{b,t}$ is the logarithm of outstanding loans to non-financial corporations over total assets, or the logarithm of holdings of government securities over total assets, for the whole balance sheet and separately by counterparty region. Standard errors are clustered at the bank level.

Table 6 reports the results. Panel A shows that exposed banks reduced their overall stock of lending to non-financial corporations after the election, and that the reduction is concentrated in the euro area portfolio: the coefficient for euro area lending is negative and significant at the 5 percent level, while the coefficients for U.S. and rest-of-world lending are not significant. Moving from the 25th to the 75th percentile of U.S. exposure implies a decline in outstanding lending to euro area firms of about 2.6 percent. The relative credit contraction identified at the loan level therefore reflects a genuine reduction in aggregate credit supply to euro area firms, not a reshuffling across borrowers. Panel B shows the counterpart on the asset side: exposed banks increased their holdings of government securities, in particular euro area securities (significant at the 10 percent level; about 3.5 percent for a 25th-to-75th percentile increase in exposure) and rest-of-world securities (significant at the 5 percent level; about 11.2 percent). Exposed banks thus shifted their asset mix away from illiquid and risky assets, such as corporate loans, toward safer and more liquid ones. Notably, the coefficient for U.S. government securities is positive but not statistically significant: in the 2024 episode, exposed banks did not rebalance toward U.S. Treasuries, consistent with the debate about the loss of the safe-haven status of U.S. assets in this period. As we show in Section 5.4, over the full sample banks exposed to U.S. EPU do rebalance toward U.S. securities as well.

4.6 Uncertainty or news about the United States?

The results so far show that banks with larger U.S. exposure tightened credit to euro area firms after the 2024 election. The central threat to interpreting this as a response to *uncertainty* is that the election also changed expected U.S. policy—on trade, taxation, regulation, and immigration—at the same time as it raised uncertainty about it. A bank with U.S. assets could then be responding to news about the expected cash flows of its U.S. borrowers or the expected value of its U.S. securities, rather than to the dispersion of outcomes around those expectations. The same concern applies, to a lesser degree, to the 2016 election. We therefore organize the remaining event-study evidence around a single falsification question: how do we know that the treatment is uncertainty rather than news about the United States? The mechanism in Section 2 yields four testable implications, which we take in turn. The panel analysis in Sections 5 and 7 returns to the same question with the continuous exposure measure.

Elections without uncertainty: the 2020 election. If exposed banks respond to changes in expected U.S. policy, any election that changes the expected policy stance should produce a contraction; if they respond to uncertainty, only elections that raise U.S. EPU should. The 2020 election changed the expected course of U.S. trade, fiscal, and regulatory policy at least as much as the 2016 election did, but it was not associated with an increase in U.S. EPU: the index had peaked with the COVID-19 shock in the spring of 2020 and was declining through the election (Figure 1). Appendix Table A3 and panel (b) of Figure 5 show that, if anything, more exposed banks *expanded* lending, lengthened maturities, and lowered rates relative to less exposed banks after the 2020 election, with wide confidence intervals in the dynamic estimates. A change in expected U.S. policy without an increase in uncertainty does not trigger the channel.

Uncertainty without an election: the 2018 trade war. Conversely, an increase in U.S. EPU that is not tied to an election should trigger it. The escalation of the U.S.–China trade war in the spring of 2018 raised U.S. EPU sharply outside any electoral cycle. Using 2017Q2–2018Q1 as the pre-period, 2018Q2 onward as the post-period, and U.S. exposure measured in 2017Q1, Appendix Table A4 shows that exposed banks reduced total new credit and term loans and raised loan rates for euro area firms after the escalation. The dynamic estimates in Appendix Figure A3 display no pre-trends and a contraction that deepens through 2019. The 2016 election, which raised U.S. EPU sharply and persistently, provides a third data point: with 2015Q4–2016Q3 as the pre-period and 2016Q4–2017Q4 as the post-period, Appendix Table A1 shows that a 25th-to-75th percentile increase in U.S. exposure predicts a decline of about 5.5 percent in total credit and 5.9 percent in term loans, with no effect on credit lines, higher loan rates (about 10 basis points), and shorter

maturities (about 3.2 percent, or 46 days). The dynamic estimates in panel (a) of Figure 5 show no pre-trends and a contraction that builds up from 2017Q1 onward. These loan-level estimates for 2016 and 2018 rely on pre-2018 originations observed retrospectively in AnaCredit and are subject to the survival selection discussed in Section 3, which can overstate a contraction by exposed banks; we read them as corroborating rather than as an independent replication. The contemporaneous bank-level evidence for 2016, drawn from supervisory reporting, is not subject to this selection, and it replicates the 2024 pattern (Appendix Table A2): exposed banks reduced outstanding lending to non-financial corporations overall and in the euro area, increased holdings of euro area government securities, and experienced an increase in the PD of their euro area borrowers. Across the four episodes, the channel is active exactly when U.S. EPU rises, whether or not an election is involved.

The sign of the news. A third implication concerns the direction of the response. If exposed banks reacted to expected U.S. fundamentals, the response should carry the sign of the news. The news that arrived after the 2024 election was, for the value of U.S. assets, favorable on average: U.S. equity prices rose to record levels in the weeks following the election, in anticipation of lower corporate taxes and lighter regulation. A bank responding to expected returns on its U.S. portfolio should therefore have expanded its risk-taking. It contracted, and it did so by tilting toward reversible instruments and away from long-term commitments, which is the signature of a response to the dispersion of outcomes rather than to their mean. The pattern of portfolio reallocation points in the same direction: in 2024 exposed banks did not rebalance toward U.S. Treasuries, as they would have if the shock had been bad news confined to U.S. private borrowers, but toward euro area and other non-U.S. government securities (Table 6), consistent with uncertainty about the U.S. policy environment as a whole.

Conditioning on U.S. fundamentals and other dimensions of U.S. risk. Finally, the panel design allows us to condition directly on the news. Appendix Table A14, Panel B, rebuilds *BEPU* from the residual of the U.S. EPU index after projecting it on real U.S. GDP growth, S&P 500 returns, the federal funds rate, and the VIX; the estimated effect is unchanged. Section 7 shows that it also survives horse races against bank exposure to U.S.–China trade tensions, U.S. monetary policy uncertainty, geopolitical risk, and the VIX, constructed with the same exposure weights. Movements in U.S. EPU that are orthogonal to U.S. macro-financial conditions and to other measures of U.S. and global risk are what drive the results.

Taken together, the four episodes indicate that the effects of U.S. elections on euro area credit are significant only when the election triggers a spike in U.S. policy uncertainty, that such spikes affect lending whether or not they coincide with an election, and that

the response is inconsistent with a reaction to first-moment news about the United States. This motivates the second step of our analysis, which exploits the full time-series variation in U.S. EPU.

5 Bank-level exposure to U.S. policy uncertainty

The event studies establish that sharp increases in U.S. policy uncertainty are transmitted to euro area credit markets by exposed banks. In this section we move beyond discrete events and exploit the full time-series variation in U.S. EPU over 2014Q4–2025Q2, using the bank-level exposure measure $BEPU_{b,t-1}$ defined in equation (2). This has two advantages. First, it uses all the variation in the data, including the many smaller movements in uncertainty between the major episodes. Second, it allows us to study a richer set of outcomes, mechanisms, and robustness checks with adequate statistical power.

5.1 Empirical specification

The baseline panel specification at the bank–firm–quarter level is

$$Y_{b,f,t} = \beta\, BEPU_{b,t-1} + \Gamma X_{b,t-1} + \alpha_{f,t} + \alpha_{b,f} + \varepsilon_{b,f,t}, \tag{7}$$

where $Y_{b,f,t}$ is the logarithm of new credit granted by bank b to firm f in quarter t (or, in later specifications, the loan rate, the logarithm of loan maturity, or a relationship-exit indicator), and the controls and fixed effects are defined as in equation (3). The interaction structure of $BEPU_{b,t-1}$ is central to identification. Because US EPU_{t-1} is common to all banks in a given quarter, the firm×quarter fixed effects absorb its direct effect on credit demand, on euro area macroeconomic conditions, and on any other common shock. Identification therefore stems from differential exposure across banks: for a given level of U.S. uncertainty, banks with higher U.S. exposure are more strongly affected, and we compare their lending to that of less exposed banks lending to the same firm in the same quarter. Bank×firm fixed effects absorb persistent relationship characteristics, and the lagged bank controls absorb time-varying differences in size, capitalization, liquidity, profitability, and asset quality. A negative β in credit regressions indicates that banks more exposed to the United States contract lending more strongly when U.S. uncertainty rises; a positive β in loan-rate or exit regressions indicates a tightening of lending conditions.

We report three sets of fixed effects throughout. The first includes bank and firm×quarter fixed effects. The second, our preferred specification, includes firm×quarter and bank×firm fixed effects. The third replaces firm×quarter with firm and industry×location×quarter fixed effects (using NACE four-digit industries), which relaxes the requirement of multiple lending relationships and roughly doubles the sample, at the cost of controlling for

demand only within industry–location cells rather than within firms (Altavilla, Canova and Ciccarelli, 2020). Standard errors are clustered at the bank–quarter level.

Threats to identification. The interaction structure of *BEPU* does not, by itself, rule out a second threat. Banks with large U.S. exposures are disproportionately large, globally active groups (Section 3 and Appendix Table A7), and their lending may be more sensitive than that of other banks to *any* global shock, not only to U.S. policy uncertainty. If so, *BEPU* would pick up the differential response of global banks to whatever comoves with U.S. EPU. We address this threat in four ways. First, the lagged bank controls, in particular size, and the bank×firm fixed effects absorb the level differences between global and other banks; what remains is the differential response of exposed banks to time variation in U.S. EPU. Second, Section 7 runs horse races in which *BEPU* enters jointly with bank exposure to U.S.–China trade tensions, U.S. monetary policy uncertainty, geopolitical risk, and the VIX, each constructed with the same exposure weights, so that a generic global-bank sensitivity would load on all of them; *BEPU* retains its effect while the alternatives lose theirs. Third, exposure measured on the liability side, which global banks also have, has no explanatory power once asset exposure is included (Section 5.6). Fourth, the event-study evidence in Section 4.6 shows that the same exposed banks did not contract after the 2020 election, which a generic sensitivity to U.S. events would not predict. A remaining limitation is that the four-quarter average share in *BEPU* is predetermined relative to lending in quarter t but not necessarily relative to the uncertainty shock, since banks could adjust their geographical exposure in anticipation of uncertainty. The negative correlation between average U.S. exposure and U.S. EPU over the sample (Section 3) argues against anticipatory build-ups in the aggregate but does not establish cross-bank exogeneity; the natural complement to this design, which we view as part of the same identification argument, interacts U.S. EPU with each bank's exposure fixed at a baseline date well before the shocks.

5.2 Credit volumes

Table 7 reports the baseline estimates for the logarithm of new total credit (columns 1–3) and new term loans (columns 4–6). Across all specifications, the coefficient on $BEPU_{b,t-1}$ is negative and statistically significant at the 5 percent level or better. Banks more exposed to U.S. policy uncertainty grant less new credit to euro area firms than less exposed banks lending to the same firm in the same quarter. The estimates are stable across fixed-effect structures, which suggests that unobserved demand is not driving the results: the coefficient in the most demanding specification with firm×quarter and bank×firm fixed effects (column 2) is, if anything, somewhat smaller in absolute value than in the specifications that rely on industry×location×quarter fixed effects.

The magnitudes are economically meaningful. Using the preferred specification, moving from the 25th to the 75th percentile of the distribution of *BEPU*—an increase of about 7.3, roughly one standard deviation—implies a reduction in new total credit of approximately 4.5 percent and in new term loans of approximately 5.4 percent. As in the event study, the coefficient on *BEPU* in credit-line regressions is small, unstable in sign, and never statistically significant (Appendix Table A20). The credit-line samples are much smaller and the estimates correspondingly less precise, so this is not a formal rejection of an equal response; but the pattern across every specification is that the contraction is concentrated in term lending, the least reversible component of the corporate portfolio, which is the margin the wait-and-see channel singles out.

Appendix Table A8 confirms the result at the bank level, where $BEPU_{b,t-1}$ is negatively and significantly related to the logarithm of total new credit and new term loans with bank and quarter fixed effects, with or without bank controls. Because the bank-level regression cannot control for credit demand, we regard the loan-level estimates as the causal benchmark; the bank-level estimates show that the relative contraction identified within firms is also visible in banks' aggregate origination.

5.3 Credit conditions: prices, maturities, and relationship termination

We next examine whether exposure to U.S. policy uncertainty affects the terms and conditions of new lending, beyond volumes. Table 8 reports estimates of (7) for three outcomes, using the same three fixed-effect structures as in Table 7.

Loan rates. Panel A uses as dependent variable the annualized interest rate agreed at origination, computed as a loan-size-weighted average across the instruments originated by the bank–firm pair in the quarter. Across all specifications, higher exposure to U.S. policy uncertainty is associated with a statistically significant increase in loan rates. In the preferred specification, moving from the 25th to the 75th percentile of *BEPU* raises annualized borrowing costs by about 90 basis points. Banks therefore respond to heightened policy uncertainty not only by contracting quantities but also by raising the price of credit, consistent with a higher risk premium demanded on irreversible commitments. This magnitude is considerably larger than the few basis points implied by the 2024 event study (Section 4.3), and the two should not be expected to coincide. The event coefficient measures the differential repricing of new loans within three quarters of a single shock, whereas the panel coefficient averages the response of loan rates over eleven years of variation in U.S. EPU, including the 2018–2019 trade war and the pandemic, over horizons at which rates have had time to adjust; the panel estimate is also more sensitive

to the high-EPU episodes that dominate the variation in *BEPU*. We regard the event estimate as a lower bound on the short-run pass-through of a single episode and the panel estimate as an average over episodes and horizons.

Loan maturity. According to the wait-and-see channel, banks facing higher uncertainty should reduce their exposure to illiquid and irreversible positions. Panel B tests whether exposed banks grant loans of shorter maturity. The dependent variable is the logarithm of the original contractual maturity of newly granted loans, measured as a loan-size-weighted average at the bank–firm–quarter level.[4] The coefficient on $BEPU_{b,t-1}$ is negative and significant across specifications: a 25th-to-75th percentile increase in exposure shortens loan maturity by about 3.4 percent, corresponding to roughly 70 days. Banks tilt lending toward contracts that are more easily reversible, thereby limiting their exposure to long-term risk while uncertainty is elevated.

Relationship termination. Does an increase in U.S. EPU also lead exposed banks to terminate pre-existing lending relationships? Panel C uses as dependent variable the indicator $Exit_{b,f,t}$, equal to one if a bank–firm relationship active in quarter $t-1$ ceases to exist in quarter t. The coefficient on $BEPU_{b,t-1}$ is positive and statistically significant across specifications: a 25th-to-75th percentile increase in exposure raises the probability of termination by about 0.8 percentage points. By contrast, the probability that a new relationship is established does not respond to *BEPU* (Appendix Table A10). Banks respond to heightened foreign policy uncertainty by actively pruning their existing loan portfolios rather than by becoming more selective at entry.

Taken together, Tables 7 and 8 show that euro area banks transmit U.S. policy uncertainty to domestic credit markets through a tightening of credit supply along every margin of the loan contract, consistent with hypothesis H2.

5.4 Bank-level outcomes: portfolio reallocation and credit risk

We next ask how exposed banks' overall balance sheets respond to U.S. EPU, using the bank-level counterpart of (7),

$$Y_{b,t} = \beta\, BEPU_{b,t-1} + \Gamma X_{b,t-1} + \alpha_t + \alpha_b + \varepsilon_{b,t}, \tag{8}$$

with standard errors clustered at the bank level. Table 9 reports the results for outstanding loans to non-financial corporations relative to total assets (Panel A) and holdings of

[4]The credit register reports original maturity in baskets (for instance 1–90 days, 90–180 days, and so on). We approximate the maturity of each instrument by the midpoint of its basket before aggregating to the bank–firm–quarter level.

government securities relative to total assets (Panel B), for the whole balance sheet and by counterparty region.

Panel A shows that higher exposure to U.S. EPU reduces the stock of corporate lending, and that the reduction is concentrated in the euro area portfolio: a one-standard-deviation increase in *BEPU* lowers total lending to non-financial corporations by about 0.6 percent and lending to euro area non-financial corporations by about 0.8 percent, the latter significant at the 5 percent level, while U.S. and rest-of-world lending do not respond significantly. Panel B shows the mirror image: the same increase in *BEPU* raises holdings of government securities by about 0.9 percent overall, with significant increases for both U.S. (1.8 percent) and euro area (1.9 percent) securities. Over the full sample, exposed banks thus substitute safe and liquid government securities for risky and illiquid corporate loans when U.S. policy uncertainty rises—a pattern that mirrors the reallocation documented for the 2024 election, with the difference that, on average over 2014–2025, the rebalancing extends to U.S. securities.

Appendix Table A9 completes the picture on the risk side, with the caveat, noted in Section 4.2, that PDs measure the expected default frequency assigned by banks' internal models rather than the dispersion of outcomes. Higher *BEPU* raises the exposure-weighted PD of borrowers in every region of the portfolio: a one-standard-deviation increase in *BEPU* raises borrower PDs by about 11.9 percent for U.S. borrowers, 4.9 percent for euro area borrowers, and 7.9 percent for rest-of-world borrowers. The largest effect is, again, on the U.S. portfolio, in line with hypothesis H1, but the increase in the riskiness of the euro area portfolio is also notable: it indicates that the tighter lending standards documented above are accompanied by a deterioration in the perceived quality of exposed banks' domestic borrowers, which reinforces the precautionary motive for reallocating toward safe assets.

5.5 Heterogeneity across banks and firms

We next examine whether the transmission of U.S. policy uncertainty varies systematically with the characteristics of banks and of their borrowers, as predicted by hypothesis H3.

Bank characteristics. A large theoretical and empirical literature emphasizes that banks' ability to absorb adverse shocks depends on their liquidity buffers, asset quality, and capital positions. These characteristics shape both funding constraints and risk-bearing capacity, and may therefore amplify or dampen the effect of heightened uncertainty on lending. Models of the bank lending channel predict stronger contractionary effects for banks with weaker balance sheets, which face tighter funding constraints and a higher marginal cost of intermediation (Bernanke and Gertler, 1995; Kashyap and Stein, 2000). Theories of precautionary behavior under uncertainty imply that banks with limited

liquidity buffers hoard liquidity and cut lending more aggressively (Berger et al., 2022). Banks with poorer asset quality may be more sensitive to uncertainty because higher expected default risk interacts with binding internal or regulatory risk constraints. We test these predictions by augmenting the baseline specification with an interaction between $BEPU_{b,t-1}$ and an indicator for a bank characteristic:

$$\log(\text{Credit})_{b,f,t} = \beta_1 \, BEPU_{b,t-1} + \beta_2 \, BEPU_{b,t-1} \times \text{Char}_b + \Gamma X_{b,t-1} + \alpha_{f,t} + \alpha_{b,f} + \varepsilon_{b,f,t}, \quad (9)$$

where Char_b identifies banks with non-performing loan ratios above the 75th percentile, excess liquidity (excess reserves over total assets) below the 25th percentile, or CET1 ratios below the 25th percentile of the cross-sectional distribution.

Columns (1)–(3) of Table 10 reveal pronounced heterogeneity. The contraction in credit supply following an increase in *BEPU* is significantly stronger for banks with high non-performing loan ratios and for banks with low excess liquidity: in both cases the interaction term is negative and significant, and the uninteracted coefficient is small and insignificant, indicating that the average effect is driven by vulnerable banks. In contrast, there is no statistically significant differential effect for banks with low capital ratios. Appendix Table A11 shows that this pattern is robust to alternative fixed-effect structures. These findings are consistent with the balance-sheet amplification channel: banks with limited liquidity buffers engage in precautionary cutbacks, and banks with weaker asset quality face tighter effective risk constraints when uncertainty rises. The interaction for low-capital banks is negative but imprecise, and we cannot reject that it equals the NPL or liquidity interactions; the evidence therefore points to asset quality and liquidity as amplifying margins rather than establishing that capital plays no role, which is consistent with capital ratios of euro area Significant Institutions being comfortably above requirements over our sample period.

Firm characteristics. We next investigate whether the transmission varies across borrowers. Lenders facing heightened uncertainty may rebalance credit away from riskier borrowers, whose default probabilities are more sensitive to adverse aggregate conditions, and away from borrowers directly exposed to the U.S. economy through trade, whose expected cash flows and perceived credit risk are affected by the same shock that impairs the bank's U.S. portfolio. We test these predictions by interacting $BEPU_{b,t-1}$ with indicators for firms with a PD above the 75th percentile, where PD is the exposure-weighted average of the internal risk assessments of the firm's lenders, and for firms operating in country–sector cells whose net exports to the United States, as defined in Section 3, lie above the 75th percentile.

Columns (4)–(5) of Table 10 show substantial heterogeneity across borrowers. The contraction associated with higher *BEPU* is significantly stronger for firms with high

ex ante default risk, and it is significantly stronger for firms in sectors that are net exporters to the United States; in both cases the uninteracted coefficient also remains negative and significant. Banks therefore reallocate credit away from the borrowers whose repayment capacity is most sensitive to U.S. policy uncertainty. As a result, international uncertainty shocks are amplified through borrower characteristics: firms with the strongest real linkages to the United States are also those that face the largest tightening of credit from their U.S.-exposed lenders.

5.6 Alternative exposure measures

Our baseline exposure measure is based on banks' direct U.S. asset holdings. Two alternative channels could generate a correlation between U.S. exposure and lending: exposure to U.S. funding, and indirect exposure through borrowers that trade with the United States. We examine both.

Liability exposure. Table 11 contrasts asset- and liability-based versions of *BEPU*, where the latter uses the share of U.S. liabilities in total liabilities from FINREP F20.06 in place of the U.S. asset share. Column (2) adds the lagged share of U.S. liabilities as a control: the coefficient on the asset-based *BEPU* is unchanged. Column (3) includes both asset- and liability-based *BEPU* jointly: the asset-based measure retains its negative and significant coefficient, while the liability-based measure is small and insignificant. Consistent with the event-study evidence in Section 4.3, the transmission operates through banks' U.S. assets rather than their U.S. funding. When entered alone, liability-based *BEPU* is also negatively related to lending (Appendix Table A12), reflecting the correlation between U.S. assets and U.S. liabilities across banks, but it has no independent explanatory power.

Borrower trade exposure. Paravisini, Rappoport and Schnabl (2023) show that the within-firm estimator can be biased when loan demand is bank-specific—for instance, when banks specialize in lending to exporters and export demand shocks are correlated with the bank characteristic under study. In our setting, banks with large U.S. asset exposures might also specialize in lending to firms that trade with the United States, whose credit demand falls when U.S. uncertainty rises. To address this concern, we construct a trade-based exposure measure that weights each bank's lending across country–sector cells by the cells' export exposure to the United States, $\alpha^{US}_{c,s,\tau} = \mathrm{EX}_{c,s\to US,\tau}/\mathrm{EX}_{c,s,\tau}$:

$$BEPU^{\text{trade}}_{b,t-1} = \frac{1}{4}\sum_{i=1}^{4} \text{TradeUSExp}_{b,t-i} \times \text{US EPU}_{t-1}, \qquad \text{TradeUSExp}_{b,\tau} = \frac{\sum_c \sum_s \alpha^{US}_{c,s,\tau}\, L_{b,c,s,\tau}}{L_{b,\tau}}, \tag{10}$$

where $L_{b,c,s,\tau}$ is bank b's lending to country–sector cell (c, s) and $L_{b,\tau}$ its total lending. We also construct a version based on net exports to the United States. Table 12 includes the trade-based measures alongside the baseline. Both trade-based measures are negatively and significantly related to new credit, consistent with a demand-side channel operating through borrowers' trade exposure. Crucially, the coefficient on the asset-based *BEPU* is unchanged, and if anything larger in absolute value, when the trade-based measures are included. The balance-sheet channel we document is therefore distinct from, and additional to, the transmission of U.S. uncertainty through borrowers' trade links (Appendix Table A13 reports the trade-based measures entered alone).

6 Real effects on euro area firms

The evidence so far shows that euro area banks exposed to the United States tighten credit supply to domestic firms when U.S. policy uncertainty rises. Whether this matters for the real economy depends on whether affected firms can substitute toward other lenders or other sources of finance. In this section we test hypothesis H4 by tracing the effects of banks' exposure to U.S. EPU on firm-level borrowing, investment, and employment.

6.1 Empirical specification

We construct a firm-level exposure measure that captures the average exposure of a firm's lenders to U.S. policy uncertainty. For each firm f and year t,

$$Exposed_{f,t-1} = \sum_{b} \omega_{b,f,t-1}\, BEPU_{b,t-1}, \tag{11}$$

where $\omega_{b,f,t-1}$ is the share of bank b in firm f's total outstanding bank borrowing in the previous year. $Exposed_{f,t-1}$ is thus high for firms that borrow predominantly from banks with large U.S. exposures in periods of high U.S. uncertainty. It has a mean of 3.3 and a standard deviation of 6.7, and moving from its 25th to its 75th percentile corresponds to an increase of about 3.3 (Table 1). We estimate

$$\log(Y_{f,t}) - \log(Y_{f,t-1}) = \beta\, Exposed_{f,t-1} + \Phi X_{f,t-1} + \alpha_f + \alpha_{i,c,t} + \varepsilon_{f,t}, \tag{12}$$

where $Y_{f,t}$ is, in turn, new bank loans, total debt, fixed assets (investment), and employment. The specification is estimated at annual frequency over 2014–2023, the period for which Orbis balance-sheet data are available.

Firm fixed effects α_f absorb time-invariant firm characteristics such as managerial quality, technology, or long-run growth potential. Industry×country×year fixed effects $\alpha_{i,c,t}$ (NACE two-digit industries) control flexibly for common demand shocks, business-

cycle conditions, and sectoral trends within each country. The vector $X_{f,t-1}$ includes the logarithm of total assets, cash over total assets, tangibility, the interest coverage ratio, and sales growth, all lagged one year. Standard errors are clustered at the firm level. Because the treatment originates at the lender level and many firms share the same lenders, firm-level clustering may understate the dependence across firms exposed to the same bank, which should be borne in mind when reading the standard errors in Table 13. Identification exploits variation in firms' exposure to U.S. policy uncertainty that arises from differences in the balance-sheet exposure of their lenders. Because lender shares are predetermined and the exposure measure varies across firms with otherwise similar observable characteristics operating in the same industry, country, and year, we interpret β as the association between lender-transmitted international uncertainty and firms' real outcomes, subject to the caveats below.

Threats to identification. Because $Exposed_{f,t-1}$ is built from the composition of a firm's lenders, the main threat is that firms that borrow from U.S.-exposed banks differ from other firms in ways that also affect their growth. Three features of the design address it. First, firm fixed effects absorb any time-invariant matching between firm types and bank types, so that identification comes from changes in a firm's exposure over time; given that lender shares are sticky at annual frequency, these changes are driven predominantly by movements in U.S. EPU rather than by switches between lenders. Second, industry×country×year fixed effects absorb the shocks most likely to be correlated with lender composition, such as sector-specific export demand or country-level cycles, and the firm controls absorb observable differences in size, liquidity, tangibility, and leverage. Third, and most importantly, the loan-level results of Section 5 establish, within the same firm and quarter, that exposed banks cut credit to a given firm relative to its other lenders. The firm-level regressions ask whether that within-firm supply shock aggregates into a change in the firm's total borrowing and real activity, which is a question about substitution rather than about the identification of the supply shock itself. The cleanest test of our claim that the channel reaches firms without direct U.S. exposure is the comparison, below, across firms with the same lender exposure but different direct trade exposure to the United States: if the results reflected borrower-side rather than lender-side exposure, they should be confined to net exporters to the United States.

6.2 Borrowing, investment, and employment

Table 13 reports the estimates. Firms whose lenders are more exposed to U.S. policy uncertainty experience a significant contraction in bank credit (column 1): moving from the 25th to the 75th percentile of the exposure distribution lowers the growth of new bank borrowing by about 7.9 percent. The contraction is not offset by other sources of

finance. Total debt growth also falls (column 2), by about 0.3 percentage points for the same increase in exposure; the smaller magnitude reflects the larger base, since total debt includes the stock of all outstanding liabilities, but the sign and significance indicate that firms do not fully substitute toward bonds, trade credit, or other lenders. The credit contraction documented at the loan level therefore translates into a meaningful reduction in firms' access to external finance. The firm and industry×country×year fixed effects absorb sector-specific shocks and time-invariant differences across firms, and the within-firm loan-level results establish the supply shock at its source; the firm-level estimates are therefore consistent with supply-side transmission through banks' balance sheets, although, as discussed above, they cannot rule out time-varying lender matching with the same force as the loan-level design.

Columns (3)–(4) show that the tightening in credit is accompanied by lower real activity. Firms with greater exposure to U.S. policy uncertainty through their lenders exhibit significantly lower investment growth: a 25th-to-75th percentile increase in exposure reduces the growth of fixed assets by about 2 percent. Employment growth also falls, by about 0.5 percent. These magnitudes are sizable relative to the average annual growth rates of investment and employment in the sample (Table 1), and they are consistent with the wait-and-see behavior of exposed banks being passed on to their borrowers' irreversible decisions.

6.3 Net exporters to the United States

The framework predicts that firms with direct trade links to the United States face a compound exposure: their own cash flows are affected by U.S. policy uncertainty, and their lenders cut credit to them disproportionately (Section 5.5). We test whether the real effects are correspondingly larger by interacting $Exposed_{f,t-1}$ with an indicator for firms in country–sector cells with positive net exports to the United States. Table 14 reports the results. The interaction is negative and significant for bank loans, total debt, and investment: relative to otherwise similar firms with the same lender exposure, net exporters to the United States experience an additional decline of about 0.7 percent in bank borrowing, 0.07 percentage points in total debt growth, and 0.5 percent in investment growth for a 25th-to-75th percentile increase in exposure. The interaction for employment is small and insignificant, suggesting that exporters adjust first along the investment margin. The uninteracted coefficients remain negative and significant, so the real effects are present for all firms borrowing from exposed banks and are amplified for those with the strongest trade linkages to the United States.

Together, these results indicate that international policy uncertainty shocks propagate beyond the financial sector and generate economically meaningful real effects. By reducing banks' willingness to supply credit, heightened U.S. policy uncertainty depresses the

investment and employment of euro area firms, transmitting foreign uncertainty to domestic economic activity through the bank lending channel.

7 Policy uncertainty versus other sources of global risk

A central concern in interpreting our results is whether the estimated effects of U.S. economic policy uncertainty merely reflect broader sources of global risk or financial-market volatility to which U.S.-exposed banks are also more sensitive. Periods of high U.S. EPU may coincide with trade tensions, uncertainty about U.S. monetary policy, geopolitical risk, or a tightening of the global financial cycle. To address this concern, we examine whether exposure to U.S. EPU continues to affect euro area credit supply after controlling for bank exposure to these alternative dimensions of risk.

7.1 Trade, monetary policy, and geopolitical uncertainty

We augment the baseline specification by jointly including bank-level exposure measures to other prominent sources of uncertainty:

$$Y_{b,f,t} = \beta_1 \, BRisk_{b,t-1} + \beta_2 \, BEPU_{b,t-1} + \Gamma X_{b,t-1} + \alpha_{f,t} + \alpha_{b,f} + \varepsilon_{b,f,t}, \tag{13}$$

where $BRisk_{b,t-1}$ is constructed exactly as $BEPU_{b,t-1}$ in equation (2) but replaces the U.S. EPU index with an alternative index. We consider three: the U.S.–China trade tension index of Rogers, Sun and Sun (2024), which captures trade-related policy uncertainty (available until 2023Q4); the U.S. monetary policy uncertainty index of Husted, Rogers and Sun (2020), which captures uncertainty about the future path of the federal funds rate (available until 2024Q2); and the geopolitical risk index of Caldara and Iacoviello (2022). In all cases the index is interacted with banks' U.S. exposure, ensuring comparability with *BEPU*. Because the three indices are correlated with U.S. EPU, this is a demanding test.

Table 15 reports the results for new total credit. Odd-numbered columns enter each *BRisk* measure alone; even-numbered columns add *BEPU*. Entered alone, exposure to trade tensions, to monetary policy uncertainty, and to geopolitical risk are each negatively related to credit supply, in line with the broader finding that uncertainty of any kind reduces lending by exposed banks. Once *BEPU* is included, however, the coefficient on $BEPU_{b,t-1}$ is negative and statistically significant in every column and of a magnitude close to the baseline, while the coefficients on trade and monetary policy uncertainty become small and insignificant, and only geopolitical risk retains an independent, smaller effect. Table 16 shows the same pattern for loan rates: *BEPU* raises loan rates significantly

in all three horse races, and the alternative measures either lose significance or retain a smaller effect. Economic policy uncertainty therefore captures a dimension of risk that is not subsumed by trade, monetary, or geopolitical uncertainty.

7.2 Policy uncertainty and the global financial cycle

A related question is whether the bank lending channel of policy uncertainty is distinct from the global financial cycle, which is commonly proxied by the VIX (Rey, 2015; Miranda-Agrippino and Rey, 2020). To answer it we broaden the geographical scope of exposure and compare two measures: $\text{BGEPU}_{b,t-1}$, which interacts banks' average exposure to countries outside the euro area over the previous four quarters with a global EPU index, computed as the GDP-PPP-weighted average of the EPU indices of fourteen non-euro-area countries; and $\text{BVIX}_{b,t-1}$, which interacts the same extra-euro-area exposure with the lagged VIX.[5] Using the same extra-euro-area exposure weights for both measures ensures that any difference between them reflects the uncertainty index rather than the composition of exposure.

Table 17 reports the results for new total credit (columns 1–3) and loan rates (columns 4–6). When entered separately, both global policy uncertainty and the VIX are associated with lower credit supply and higher loan rates by exposed banks. When the two measures enter jointly, global policy uncertainty remains associated with lower credit and higher loan rates, whereas the VIX coefficients shrink and become statistically insignificant. The bank lending channel of policy uncertainty is thus distinct from, and empirically dominates, the transmission of global financial-market volatility through exposed banks.

Taken together, the results in this section indicate that the contraction in euro area credit supply documented in Sections 4 and 5 reflects a specific policy-uncertainty channel operating through internationally exposed banks, rather than a generic response to global risk. This is consistent with the observation in Figure 1 that policy uncertainty spikes around politically consequential events that are not necessarily associated with financial distress, and with the wait-and-see mechanism, in which it is uncertainty about future policy, rather than current financial conditions, that raises the value of postponing irreversible commitments.

8 Robustness

This section summarizes a battery of checks on the baseline panel estimates of Section 5. The corresponding tables are reported in the Appendix. Throughout, we report the three

[5]The fourteen countries are Australia, Brazil, Canada, Chile, China, India, Japan, Korea, Mexico, Pakistan, Russia, Sweden, the United Kingdom, and the United States.

fixed-effect structures of Table 7.

Level of aggregation. Our baseline regressions aggregate instruments to the bank–firm–quarter level. Appendix Table A19 re-estimates the baseline at the loan level, with more than ten million observations. The coefficients on $BEPU_{b,t-1}$ remain negative and statistically significant for both total credit and term loans, with magnitudes about half of the bank–firm-level estimates, reflecting the different weighting of small and large instruments. The contraction in credit supply is not an artifact of aggregation across lending relationships.

The COVID-19 period. U.S. EPU reached exceptional levels during the pandemic, and lending in 2020–2021 was shaped by public guarantee schemes and extraordinary monetary and supervisory measures. Panel A of Appendix Table A14 excludes the period 2019Q4–2021Q1. The estimated coefficients are similar in magnitude to the baseline and more precisely estimated, so the results are not driven by pandemic-specific dynamics.

U.S. macro-financial conditions. Movements in U.S. EPU may be correlated with U.S. business-cycle conditions, equity returns, monetary policy, and financial volatility, to which U.S.-exposed banks are also sensitive. Panel B of Appendix Table A14 replaces the U.S. EPU index with the residual from a regression of the index on real U.S. GDP growth, S&P 500 returns, the federal funds rate, and the VIX, and rebuilds *BEPU* using this residualized index. The coefficients are essentially unchanged: the effects are not driven by contemporaneous macroeconomic or financial conditions in the United States.

Domestic policy uncertainty. U.S. and euro area policy uncertainty comove (Appendix Figure A1), and banks with large U.S. exposures may also be more sensitive to uncertainty at home. Appendix Table A15 adds a domestic EPU measure assigned according to the country of each bank's headquarters, available for banks headquartered in Germany, France, Greece, Ireland, Italy, and Spain. The coefficient on $BEPU_{b,t-1}$ remains negative and significant, and its magnitude is unchanged; domestic EPU enters with small coefficients of unstable sign. Foreign policy uncertainty affects euro area credit supply over and above domestic uncertainty.

Timing of U.S. EPU. The baseline uses the lagged quarterly average of the monthly EPU index. Appendix Table A16 shows that the results are robust to using the contemporaneous quarterly average (Panel B) or the lagged end-of-quarter value (Panel C), with coefficients of very similar magnitude across timing conventions.

Alternative asset-based exposure measures. Our baseline U.S. exposure includes all asset classes vis-à-vis U.S. counterparties. Appendix Table A17 rebuilds *BEPU* using only total loans and advances to U.S. counterparties (columns 1–3) or only loans and advances to U.S. non-financial corporations (columns 4–6). The coefficients are negative and significant in all cases, and larger in absolute value for the narrower NFC-loan measure, consistent with the credit-risk interpretation of the first stage in Section 4.2.

Additional bank controls. Appendix Table A18 reports the coefficients on the bank controls and adds the non-performing loan ratio and cash over total assets to the control set. The coefficient on $BEPU_{b,t-1}$ is stable. Among the controls, only bank size is significantly related to new lending.

Credit lines. As discussed in Section 5.2, Appendix Table A20 shows that *BEPU* has no significant effect on the volume of credit lines under any fixed-effect structure, consistent with the interpretation that banks adjust the least reversible component of their portfolio.

Dollar-denominated loans. A natural question is whether exposed banks also cut lending in U.S. dollars, which may be more directly affected by uncertainty about U.S. policy. Appendix Table A21 restricts the sample to term loans denominated in U.S. dollars, extended to euro area and non-euro-area firms through direct cross-border lending or foreign branches, and controls for floating-rate and secured-loan indicators. The coefficient on $BEPU_{b,t-1}$ is negative and significant: a one-standard-deviation increase in exposure reduces new dollar term lending by about 2.3 percent. The contraction is thus not confined to euro-denominated lending to domestic firms.

Entry. Finally, Appendix Table A10 shows that *BEPU* has no effect on the probability that a new bank–firm relationship is established, in contrast to its significant effect on relationship termination. Exposed banks adjust existing relationships rather than the flow of new ones.

Overall, the baseline findings are not sensitive to the level of aggregation, to the measurement and timing of uncertainty, to the definition of exposure, to domestic conditions, to exceptional macroeconomic episodes, or to the set of controls.

9 Conclusion

This paper asks whether uncertainty about the economic policy of a globally dominant country remains confined to that country when foreign banks hold its assets. It does

not. Combining supervisory data on the geographical composition of euro area banks' assets with the euro area credit register and firm-level balance sheets, we show that euro area banks with larger exposures to the United States tighten credit supply to domestic firms when uncertainty about U.S. economic policy rises. Within the same firm and quarter, exposed banks grant less new credit, charge higher rates, shorten maturities, and terminate more lending relationships; at the bank level, they reduce corporate lending and reallocate toward government securities. The effects are present around the 2016 and 2024 U.S. elections and the 2018 trade war, all of which raised U.S. policy uncertainty, and absent around the 2020 election, which did not. They are robust to controlling for credit demand, for U.S. macro-financial conditions, for domestic policy uncertainty, and for other sources of global risk, and they are distinct from the global financial cycle as captured by the VIX.

The evidence points to two complementary mechanisms, and different pieces of it speak to each. The signatures of a *wait-and-see* response to uncertainty are the shortening of maturities, the concentration of the contraction in term loans rather than in shorter and renewable credit lines, the reallocation toward liquid government securities in the absence of realized losses, and the absence of any contraction when expected U.S. policy changed but uncertainty did not. The signatures of *balance-sheet amplification* are the concentration of the contraction in banks with weak asset quality and thin liquidity buffers and its direction toward riskier borrowers and borrowers whose cash flows depend on trade with the United States; these patterns are consistent with ordinary risk-capacity constraints and show where the uncertainty shock bites hardest rather than that it is an uncertainty shock. We do not find that capital ratios moderate the effect, although that estimate is imprecise. Firms whose lenders are more exposed to U.S. policy uncertainty subsequently borrow less, do not substitute toward other sources of finance, and invest and hire less, with larger effects for net exporters to the United States; these firm-level patterns are consistent with the within-firm supply shock reaching the real economy.

Our findings have implications for the analysis of international spillovers and for policy. First, they identify a channel of cross-border transmission that operates neither through trade nor through asset prices, but through the internal capital allocation of global banks in response to uncertainty about foreign policy. Because this channel works through banks' foreign *asset* exposures rather than their foreign funding, it is active even when banks' liquidity positions are comfortable, and it is triggered by events—elections, trade-policy announcements—that need not be accompanied by financial distress. Second, the heterogeneity of the response implies that foreign policy uncertainty interacts with domestic financial vulnerabilities: the same shock produces a larger credit contraction where asset quality is weaker, which is relevant for macroprudential oversight of internationally active banks. Third, the concentration of the contraction on riskier and more trade-exposed

borrowers implies that U.S. policy uncertainty affects the allocation, and not only the volume, of euro area credit, compounding the direct effects of U.S. policy on firms that trade with the United States.

What is general in these findings, and what is specific to the United States? The magnitudes are specific: the United States is the largest single foreign counterparty of euro area banks, and U.S. policy uncertainty in our sample reached levels without precedent. The mechanism is not. It requires only that intermediaries hold foreign assets whose risk distribution depends on the source country's policy, that domestic lending be costly to reverse, and that risk-bearing capacity be constrained. Under these conditions, policy uncertainty in any country large enough to matter for the asset side of foreign banks should be transmitted to credit conditions in the countries where those banks lend. Gross foreign asset positions, in other words, create an externality through which one country's policy uncertainty shapes the allocation of credit in another.

Two extensions are natural. The first is to quantify the aggregate cost of the channel by benchmarking observed credit growth to euro area firms after each uncertainty episode against a counterfactual in which exposed banks had behaved as their unexposed peers, and by tracing the implied effect on aggregate investment. The second is to examine whether the channel operates symmetrically for other source countries and for other types of policy uncertainty, and whether central bank liquidity provision or macroprudential buffers can attenuate it. We leave these questions for future research.

Data and code availability

The loan-level data (AnaCredit) and the supervisory data (FINREP and COREP) used in this paper are confidential statistical information collected by the European System of Central Banks and the Single Supervisory Mechanism. They cannot be shared by the authors or deposited in a public repository. Access for research purposes is possible for researchers affiliated with, or visiting, the ECB or a national central bank of the Eurosystem, subject to the applicable confidentiality framework; the corresponding author can provide details on the access procedure. Firm-level balance-sheet data are from Bureau van Dijk's Orbis and are available under a commercial license. All other inputs are public: the U.S. and country-level economic policy uncertainty indices, the U.S.–China trade tension index, the U.S. monetary policy uncertainty index, the geopolitical risk index, the VIX, and the OECD Inter-Country Input-Output tables. Upon publication, we will deposit the code that constructs every variable and estimates every table and figure in the paper, the public input data, a data dictionary, and step-by-step instructions that allow an authorized researcher with access to AnaCredit and FINREP/COREP to reproduce all results.

References

Altavilla, C., M. Boucinha, and J.-L. Peydró (2018). Monetary policy and bank profitability in a low interest rate environment. *Economic Policy 33*(96), 531–586.

Altavilla, C., F. Canova, and M. Ciccarelli (2020). Mending the broken link: Heterogeneous bank lending and monetary policy pass-through. *Journal of Monetary Economics 110*, 81–98.

Baker, S. R., N. Bloom, and S. J. Davis (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics 131*(4), 1593–1636.

Berger, A., O. Guedhami, H. Kim, and X. Li (2022). Economic policy uncertainty and bank liquidity hoarding. *Journal of Financial Intermediation 49*, 100893.

Bernanke, B. S. (1983). Irreversibility, uncertainty, and cyclical investment. *Quarterly Journal of Economics 98*(1), 85–106.

Bernanke, B. S. and M. Gertler (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives 9*(4), 27–48.

Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica 77*, 623–685.

Bloom, N. (2014). Fluctuations in uncertainty. *Journal of Economic Perspectives 28*(2), 153–176.

Boehm, C. and T. Kroner (2026). The u.s., economic news, and the global financial cycle. *Review of Economic Studies 93*(1), 215–249.

Bruno, V. and H. S. Shin (2015). Capital flows and the risk-taking channel of monetary policy. *Journal of Monetary Economics 71*, 119–132.

Buch, C., M. Buchholz, and L. Tonzer (2015). Uncertainty, bank lending, and bank-level heterogeneity. *IMF Economic Review 63*, 919–954.

Caldara, D. and M. Iacoviello (2022). Measuring geopolitical risk. *American Economic Review 112*(4), 1194–1225.

Carpinelli, L. and M. Crosignani (2021). The design and transmission of central bank liquidity provisions. *Journal of Financial Economics 141*(1), 27–47.

Cetorelli, N. and L. S. Goldberg (2011). Global banks and international shock transmission: Evidence from the crisis. *IMF Economic Review 59*(1), 41–76.

Chodorow-Reich, G. (2014). The employment effects of credit market disruptions: Firm-level evidence from the 2008–9 financial crisis. *Quarterly Journal of Economics 129*(1), 1–59.

Dedola, L., G. Rivolta, and L. Stracca (2017). If the fed sneezes, who catches a cold? *Journal of International Economics 108*(1), S23–S41.

di Giovanni, J., Kalemli-Özcan, M. F. Ulu, and Y. S. Baskaya (2022). International spillovers and local credit cycles. *The Review of Economic Studies 89*(2), 733–773.

Giannetti, M. and L. Laeven (2012). The flight home effect: Evidence from the syndicated loan market during financial crises. *Journal of Financial Economics 104*(1), 23–43.

Gissler, S., J. Oldfather, and D. Ruffino (2016). Lending on hold: Regulatory uncertainty and bank lending standards. *Journal of Monetary Economics 81*, 89–101.

Gulen, H. and M. Ion (2016). Policy uncertainty and corporate investment. *Review of Financial Studies 29*(3), 523–564.

Husted, L., J. Rogers, and B. Sun (2020). Monetary policy uncertainty. *Journal of Monetary Economics 115*, 20–36.

Jasova, M., C. Mendicino, and D. Supera (2021). Policy uncertainty, lender of last resort and bank lending. *Journal of Monetary Economics 120*, 78–98.

Jiménez, G., S. Ongena, J.-L. Peydró, and J. Saurina (2012). Credit supply and monetary policy: Identifying the bank balance-sheet channel with loan applications. *American Economic Review 102*(5), 2301–2326.

Kashyap, A. K. and J. C. Stein (2000). What do a million observations on banks say about the transmission of monetary policy? *American Economic Review 90*(3), 407–428.

Khwaja, A. I. and A. Mian (2008). Tracing the impact of bank liquidity shocks: Evidence from an emerging market. *American Economic Review 98*(4), 1413–42.

Miranda-Agrippino, S. and H. Rey (2020). U.S. Monetary Policy and the Global Financial Cycle. *The Review of Economic Studies 87*(6), 2754–2776.

Ongena, S., J.-L. Peydró, and N. van Horen (2015). Shocks abroad, pain at home? bank-firm level evidence on financial contagion during the recent financial crisis. *IMF Economic Review 63*(4), 698–750.

Paravisini, D., V. Rappoport, and P. Schnabl (2023). Specialization in bank lending: Evidence from exporting firms. *Journal of Finance 78*(4), 2049–2085.

Peek, J. and E. S. Rosengren (1997). The international transmission of financial shocks: The case of japan. *American Economic Review 87*(4), 495–505.

Peek, J. and E. S. Rosengren (2000). Collateral damage: Effects of the japanese bank crisis on real activity in the united states. *American Economic Review 90*(1), 30–45.

Popov, A. and G. F. Udell (2012). Cross-border banking, credit access, and the financial crisis. *Journal of International Economics 87*(1), 147–161.

Rey, H. (2015). Dilemma not trilemma: The global financial cycle and monetary policy independence. NBER Working Paper 21162, National Bureau of Economic Research.

Rodrik, D. (1991). Policy uncertainty and private investment in developing countries. *Journal of Development Economics 36*(2), 229–242.

Rogers, J. H., B. Sun, and T. Sun (2024). U.s.–china tension. Technical report, Working Paper.

Schnabl, P. (2012). The international transmission of bank liquidity shocks: Evidence from an emerging market. *Journal of Finance 67*(3), 897–932.

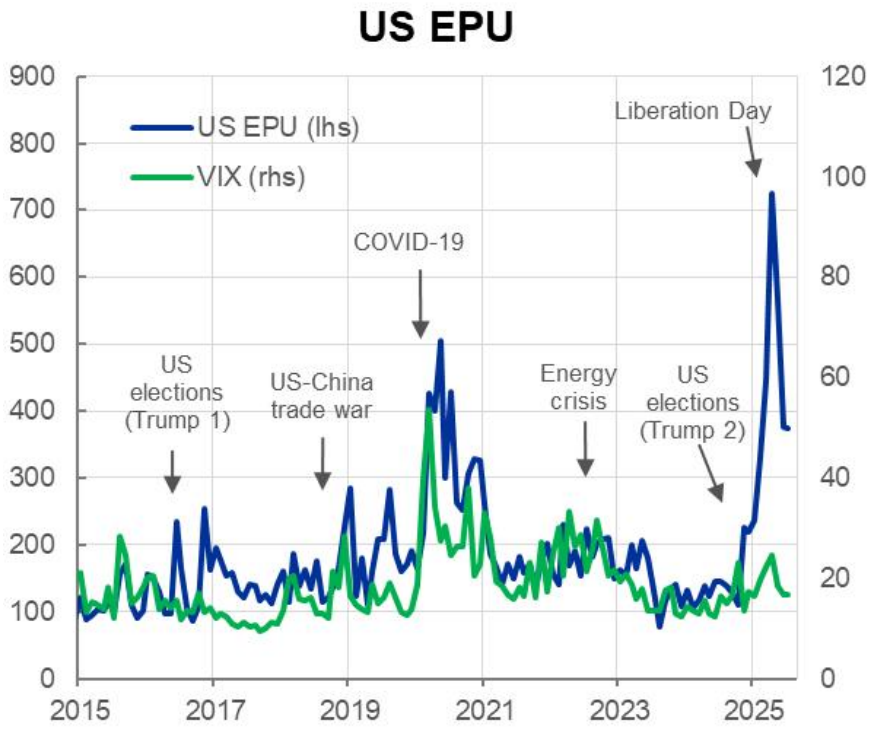
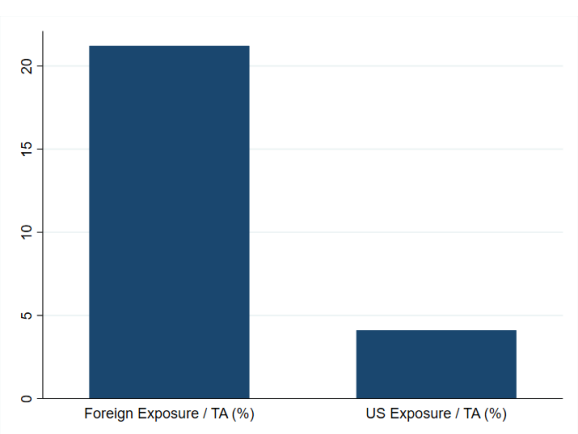
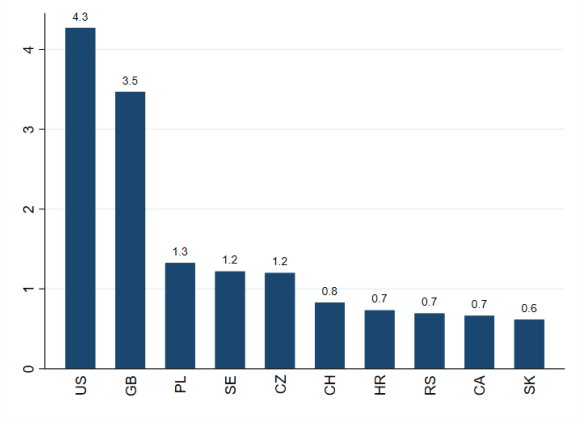


Figure 1: U.S. economic policy uncertainty and the VIX

Notes: The figure plots the monthly U.S. economic policy uncertainty (EPU) index of Baker, Bloom and Davis (2016) (left axis) and the VIX (right axis) between January 2015 and June 2025. The EPU index tracks the frequency of articles in ten leading U.S. newspapers containing terms related to "uncertainty," "economy," and "policy." Annotated events: the 2016 U.S. election, the 2018–2019 U.S.–China trade war, COVID-19, the 2022 energy crisis, the 2024 U.S. election, and the April 2025 "Liberation Day" tariff announcements.



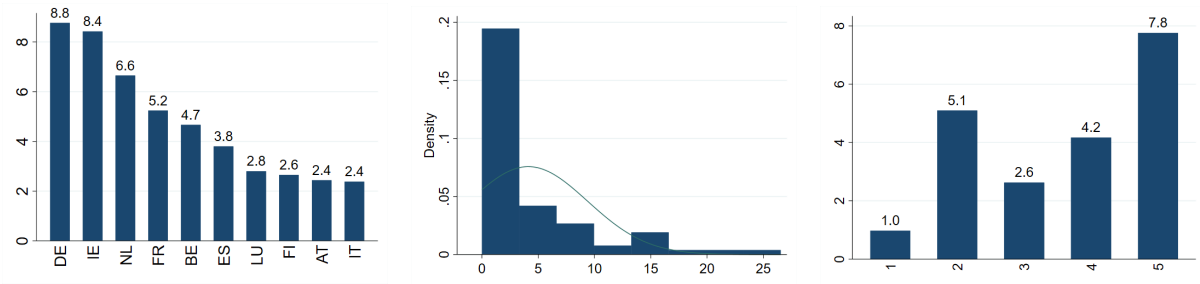
(a) Foreign and U.S. exposure



(b) Largest extra-euro-area counterparties

Figure 2: International exposures of euro area banks

Notes: Panel (a) reports the average exposure of 79 euro area banking groups to all extra-euro-area countries and to the United States, in percent of total assets, in 2024Q4. Panel (b) reports the average exposure to the ten largest extra-euro-area counterparty countries, in percent of total assets. Exposure is defined as the sum of derivatives, equity instruments, cash balances at central banks and other demand deposits, debt securities, and loans and advances vis-à-vis counterparties resident in the country. Source: FINREP F20.04.



(a) By country of headquarters **(b) Distribution across banks** **(c) By quintile of total assets**

Figure 3: Heterogeneity in U.S. exposure across euro area banks

Notes: The figure reports U.S. exposure in percent of total assets for 79 euro area banking groups in 2024Q4. Panel (a) shows the average by country of the group's headquarters; panel (b) shows the cross-sectional distribution (histogram and fitted normal density); panel (c) shows the average by quintile of total assets. Source: FINREP F20.04.

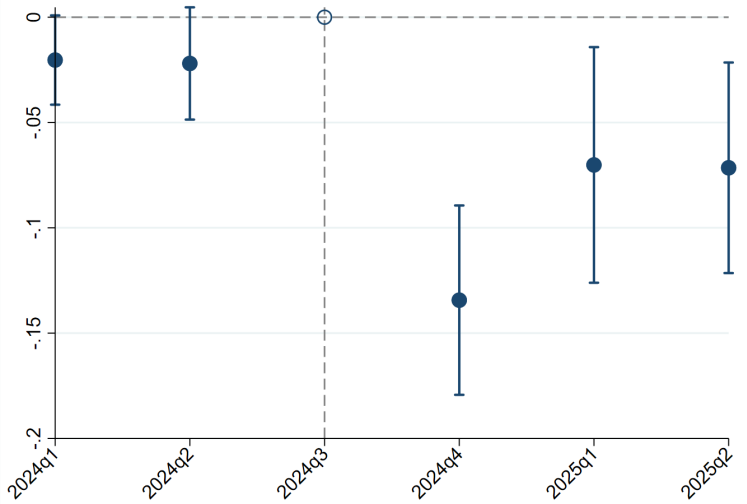
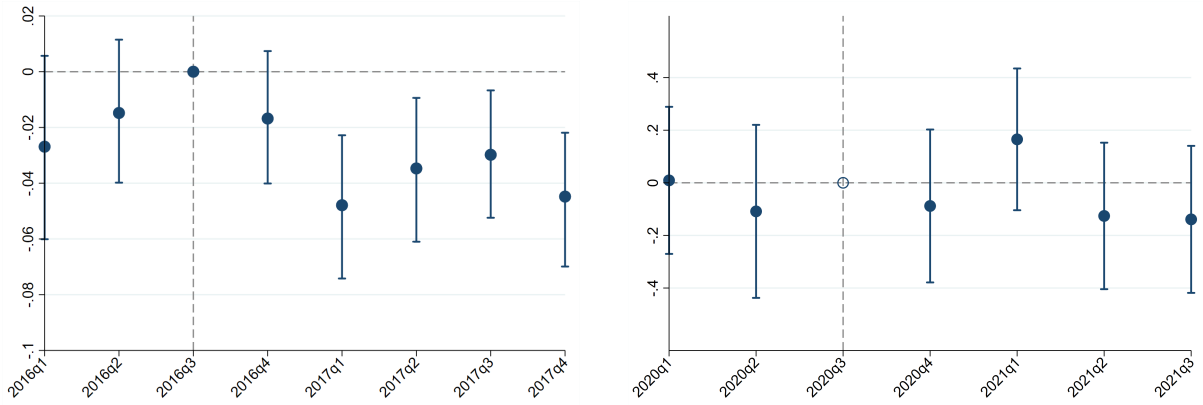


Figure 4: Dynamic effects of the 2024 U.S. election on lending to euro area firms

Notes: The figure reports the coefficients β_k from the event-study specification $\log(\text{Credit})_{b,f,t} = \sum_{k \neq 2024Q3} \beta_k \, \text{US Exp}_b \times \mathbb{1}\{t = k\} + \Gamma X_{b,t-1} + \alpha_{f,t} + \alpha_b + \epsilon_{b,f,t}$, with 90 percent confidence intervals. The dependent variable is the logarithm of total new credit granted by bank b to firm f in quarter t. US Exp_b is U.S. assets over total assets as of 2023Q3. The omitted period is 2024Q3, the quarter preceding the election, so that coefficients are interpreted relative to that quarter. The specification includes firm×quarter and bank fixed effects and lagged bank controls (log total assets, CET1 ratio, LCR, ROA, NPL ratio). Standard errors are clustered at the bank level.



(a) 2016 election **(b) 2020 election**

Figure 5: Dynamic effects of the 2016 and 2020 U.S. elections on lending to euro area firms

Notes: The figure reports event-study coefficients, with 90 percent confidence intervals, from the specification in the notes to Figure 4 applied to the 2016 election (panel a; omitted quarter 2016Q3; U.S. exposure measured in 2015Q3) and the 2020 election (panel b; omitted quarter 2020Q3; U.S. exposure measured in 2019Q3). The dependent variable is the logarithm of total new credit. The specifications include firm×quarter and bank fixed effects and lagged bank controls. Standard errors are clustered at the bank level. Note the different scales of the vertical axes.

Table 1: Descriptive statistics

Variable	*N*	Mean	SD	Min	p25	p75	Max
Panel A: Bank level (bank–quarter)							
US Exposure (US assets / total assets, %)	2,894	3.75	4.81	0.00	0.49	5.02	27.59
US Liabilities / total assets (%)	2,895	2.56	6.41	0.00	0.06	2.20	70.49
BEPU	2,499	5.78	7.41	0.00	0.81	8.11	62.86
Total assets (log)	2,859	25.29	1.62	21.20	25.12	26.38	28.59
CET1 ratio (%)	2,857	17.00	9.00	7.00	15.00	18.00	103.00
ROA (%)	2,582	0.43	0.39	0.00	0.31	0.59	1.91
NPL ratio (%)	1,358	4.84	5.66	0.15	2.13	4.82	47.90
Excess liquidity / total assets (%)	1,969	9.27	7.00	−0.24	3.54	13.87	35.75
Liquidity coverage ratio	2,894	2.01	1.13	1.14	1.81	2.02	11.64
BRisk: US–China trade tension	2,264	4.93	6.08	0.00	0.74	7.05	51.66
BRisk: Geopolitical risk	2,499	3.52	4.69	0.00	0.49	4.72	54.28
BRisk: US monetary policy uncertainty	2,421	5.85	7.54	0.00	0.80	7.98	52.52
Panel B: Bank–firm level (bank–firm–quarter)							
Total new credit (log)	23,992,315	11.17	1.78	6.11	10.12	12.20	16.23
Term loans (log)	23,592,315	11.00	1.89	4.60	9.97	12.10	16.09
Credit lines (log)	7,420,836	3.48	8.97	−27.03	−6.69	11.07	14.22
Loan rate	21,892,315	0.04	0.07	0.00	0.01	0.05	1.00
Loan maturity (log days)	22,792,315	7.16	1.39	0.00	6.28	7.97	20.55
Exit indicator	2,400,315	0.40	0.49	0.00	0.00	1.00	1.00
Borrower probability of default	23,892,315	0.07	2.43	0.00	0.00	0.02	1.00
Panel C: Firm level (firm–year)							
Exposed	5,961,893	3.27	6.67	0.00	0.00	3.25	49.63
Δ Total debt	3,212,729	−0.01	0.78	−3.63	−0.16	0.10	4.16
Δ Bank loans	3,212,729	2.98	11.04	−1.00	−0.53	1.64	85.59
Δ Investment (fixed assets)	5,192,540	0.93	4.08	−0.77	−0.07	0.34	33.41
Δ Employment	3,544,793	0.04	0.38	−12.32	0.00	0.12	12.42
Net exports to US (sector–country)	8,752,322	0.17	0.85	0.00	0.00	0.14	111.40
Total assets (log)	2,597,632	13.58	1.60	6.76	12.49	14.52	18.58
Tangibility	2,557,475	0.30	0.27	0.00	0.07	0.49	0.97
Sales growth	1,519,486	0.06	0.60	−15.81	−0.10	0.22	15.02
Cash / total assets	2,459,710	0.15	0.17	0.00	0.02	0.22	0.80

Notes: The table reports descriptive statistics for bank-level variables (Panel A), bank–firm–quarter lending outcomes (Panel B), and firm-level variables (Panel C). *BEPU* and *BRisk* are defined in Section 3.3 and equation (13). *Exposed* is the borrowing-share-weighted average *BEPU* of a firm's lenders (equation (11)). Δ denotes annual log changes. All other variables are defined in Section 3.

Table 2: 2024 U.S. election: effects on the riskiness of banks' loan portfolios

Dependent variable: $\log(\text{PD})_{b,c,t}$	(1) **Total**	(2) **US**	(3) **EA**	(4) **RoW**
Post × High US Exp_b	0.459*	0.591***	0.260	0.0935
	(0.270)	(0.182)	(0.394)	(0.660)
Observations	783	260	273	248
R^2	0.291	0.892	0.265	0.125
Quarter FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Country FE	Yes	No	No	No
Bank controls	Yes	Yes	Yes	Yes

Notes: The table reports difference-in-differences estimates of equation (4). The dependent variable is the logarithm of the exposure-weighted average probability of default (PD) of bank b's borrowers in country c in quarter t. The pre-election period is 2023Q4–2024Q3 and the post-election period is 2024Q4–2025Q2. Post_t equals one in the post-election period. High US Exp_b equals one for banks whose U.S. exposure (U.S. assets over total assets) in 2023Q3 lies above the 75th percentile of the pre-treatment distribution. Column (1) pools all borrower countries; columns (2)–(4) restrict the sample to borrowers in the United States, the euro area, and the rest of the world, respectively. All specifications include quarter and bank fixed effects and lagged bank controls; column (1) also includes borrower-country fixed effects. Standard errors, in parentheses, are clustered at the bank level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 3: 2024 U.S. election: effects on bank lending and credit conditions in the euro area

	Bank lending			**Credit conditions**		
	(1) Total credit	(2) Term loans	(3) Credit lines	(4) Exit	(5) Loan maturity	(6) Loan rate
Post × US Exp_b	−0.0769***	−0.0746***	0.0075	0.0146**	−0.0197**	0.000277**
	(0.0174)	(0.0187)	(0.0870)	(0.00672)	(0.00926)	(0.000125)
Observations	52,433	51,901	6,779	52,433	97,329	355,923
R^2	0.603	0.563	0.853	0.375	0.504	0.433
Firm × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports difference-in-differences estimates of equation (3) for lending by euro area banks to euro area firms around the 2024 U.S. election. The pre-election period is 2023Q4–2024Q3 and the post-election period is 2024Q4–2025Q2. Post_t equals one in the post-election period. US Exp_b is U.S. assets over total assets as of 2023Q3. The dependent variables are the logarithm of new total credit, new term loans, and new credit lines (columns 1–3); an indicator for the termination of the bank–firm relationship (column 4); the logarithm of loan maturity (column 5); and the annualized loan rate (column 6). Loan maturity and the loan rate are loan-size-weighted averages at the bank–firm–quarter level; the corresponding regressions additionally control for loan characteristics. All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls (log total assets, CET1 ratio, LCR, ROA, NPL ratio). Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 4: 2024 U.S. election: asset-side versus liability-side exposure

Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)
Post × US Assets/TA_b	-0.0769^{***}	-0.0391^{*}	
	(0.0174)	(0.0235)	
Post × US Liabilities/TA_b			0.00254
			(0.00682)
US Liabilities/$\text{TA}_{b,t-1}$		-0.132^{***}	
		(0.0402)	
Observations	52,433	52,433	52,433
R^2	0.630	0.630	0.630
Firm × Quarter FE	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes

Notes: The table compares the role of banks' U.S. asset and U.S. liability exposure in the transmission of the 2024 U.S. election shock to lending to euro area firms. The dependent variable is the logarithm of new total credit. Post_t equals one in 2024Q4–2025Q2. US Assets/TA_b and US Liabilities/TA_b are U.S. assets and U.S. liabilities over total assets as of 2023Q3 (FINREP F20.04 and F20.06). Column (1) is the baseline asset-based specification; column (2) adds the lagged, time-varying U.S. liability share as a control; column (3) uses the liability share as the treatment variable. All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 5: 2024 U.S. election: heterogeneity by bank and firm characteristics

	Bank characteristics		**Firm characteristics**	
	(1)	(2)	(3)	(4)
Dependent variable: $\log(\text{Total credit})_{b,f,t}$	High NPL	Low capital	High PD	High US exports
Post × US Exp_b × Char	-0.0826^{**}	-0.0293	-0.0144^{*}	-0.0045^{*}
	(0.0361)	(0.0543)	(0.0084)	(0.0019)
Post × US Exp_b	-0.0022	-0.0705^{**}	-0.0008	0.0013
	(0.0046)	(0.0336)	(0.0028)	(0.0020)
Observations	196,516	51,077	164,653	398,894
R^2	0.651	0.619	0.655	0.663
Firm × Quarter FE	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes

Notes: The table reports difference-in-differences estimates of equation (3) augmented with a triple interaction between $\text{Post}_t \times \text{US Exp}_b$ and an indicator for a bank or firm characteristic. The dependent variable is the logarithm of new total credit. US Exp_b is U.S. assets over total assets as of 2023Q3. High NPL identifies banks with NPL ratios above the 75th percentile; Low capital identifies banks with CET1 ratios below the 25th percentile; High PD identifies firms with a probability of default above the 75th percentile; High US exports identifies firms in sectors with export exposure to the United States above the 95th percentile. Lower-order interactions are absorbed by the fixed effects or included. All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 6: 2024 U.S. election: effects on bank balance-sheet composition

	(1) **Total**	(2) **US**	(3) **EA**	(4) **RoW**
Panel A: Outstanding loans to non-financial corporations / total assets (log)				
Post × US Exp_b	−0.00349*	−0.00229	−0.00400**	0.000344
	(0.00190)	(0.00866)	(0.00158)	(0.00667)
Observations	460	371	460	448
Adj. R^2	0.997	0.991	0.997	0.987
Panel B: Government securities / total assets (log)				
Post × US Exp_b	0.00394	0.0121	0.00544*	0.0173**
	(0.00263)	(0.0112)	(0.00295)	(0.00781)
Observations	594	416	594	535
Adj. R^2	0.992	0.963	0.990	0.976
Quarter FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes

Notes: The table reports bank-level difference-in-differences estimates of equation (6) around the 2024 U.S. election. The dependent variable is the logarithm of outstanding loans to non-financial corporations over total assets (Panel A) and the logarithm of holdings of government securities over total assets (Panel B), for the whole balance sheet (column 1) and by counterparty region (columns 2–4). Post_t equals one in 2024Q4–2025Q2. US Exp_b is U.S. assets over total assets as of 2023Q3. All specifications include quarter and bank fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 7: Bank exposure to U.S. policy uncertainty and new lending to euro area firms

	Total credit			**Term loans**		
	(1)	(2)	(3)	(4)	(5)	(6)
$BEPU_{b,t-1}$	−0.00907***	−0.00617***	−0.00790**	−0.0105***	−0.00740***	−0.00948**
	(0.00300)	(0.00221)	(0.00311)	(0.00346)	(0.00275)	(0.00367)
Observations	671,477	599,637	1,070,536	666,713	595,831	1,059,352
R^2	0.627	0.746	0.702	0.611	0.739	0.694
Number of banks	296	227	301	296	227	301
Number of firms	58,646	32,995	112,401	58,646	32,995	112,401
Bank FE	Yes		Yes	Yes		Yes
Firm FE			Yes			Yes
Firm × Quarter FE	Yes	Yes		Yes	Yes	
Bank × Firm FE		Yes			Yes	
Industry × Location × Quarter FE			Yes			Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimates of equation (7). The dependent variable is the logarithm of new total credit (columns 1–3) and new term loans (columns 4–6) granted by bank b to firm f in quarter t. $BEPU_{b,t-1}$ is defined in equation (2). Industry fixed effects are at the NACE four-digit level. All specifications include lagged bank controls (log total assets, CET1 ratio, LCR, ROA, NPL ratio). The sample period is 2014Q4–2025Q2. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 8: Bank exposure to U.S. policy uncertainty and credit conditions

	(1)	(2)	(3)
Panel A: Loan rate			
$BEPU_{b,t-1}$	0.00119***	0.00125***	0.000889***
	(0.000299)	(0.000312)	(0.000287)
Observations	538,145	473,006	841,647
R^2	0.317	0.609	0.384
Panel B: Loan maturity (log)			
$BEPU_{b,t-1}$	−0.00606***	−0.00461**	−0.00396**
	(0.00184)	(0.00190)	(0.00188)
Observations	649,066	580,212	1,030,717
R^2	0.471	0.633	0.567
Panel C: Exit			
$BEPU_{b,t-1}$	0.00114**	0.00110**	0.00119**
	(0.000524)	(0.000429)	(0.000495)
Observations	602,481	541,598	971,213
R^2	0.627	0.716	0.599
Bank FE	Yes		Yes
Firm FE			Yes
Firm × Quarter FE	Yes	Yes	
Bank × Firm FE		Yes	
Industry × Location × Quarter FE			Yes
Bank controls	Yes	Yes	Yes

Notes: The table reports estimates of equation (7) for the terms of new lending. In Panel A, the dependent variable is the annualized agreed loan rate at the bank–firm–quarter level, computed as the loan-size-weighted average rate across instruments originated in the quarter. In Panel B, it is the logarithm of original loan maturity, computed as the loan-size-weighted average maturity. In Panel C, *Exit* is an indicator equal to one if a bank–firm relationship existed in quarter $t-1$ and ceased to exist in quarter t. $BEPU_{b,t-1}$ is defined in equation (2). Industry fixed effects are at the NACE four-digit level. All specifications include lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 9: Bank exposure to U.S. policy uncertainty and bank balance-sheet composition

	(1) **Total**	(2) **US**	(3) **EA**	(4) **RoW**
Panel A: Outstanding loans to non-financial corporations / total assets (log)				
$BEPU_{b,t-1}$	−0.000868*	0.00184	−0.00113**	0.000956
	(0.000504)	(0.00149)	(0.000554)	(0.00225)
Observations	2,347	1,879	2,347	2,310
Adj. R^2	0.976	0.950	0.973	0.935
Panel B: Government securities / total assets (log)				
$BEPU_{b,t-1}$	0.00119***	0.00242**	0.00257***	0.00220
	(0.000407)	(0.00111)	(0.000601)	(0.00144)
Observations	2,361	1,520	2,361	2,105
Adj. R^2	0.973	0.904	0.941	0.872
Quarter FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes

Notes: The table reports bank-level estimates of equation (8) over 2014Q4–2025Q2. The dependent variable is the logarithm of outstanding loans to non-financial corporations over total assets (Panel A) and the logarithm of holdings of government securities over total assets (Panel B), for the whole balance sheet (column 1) and by counterparty region (columns 2–4). $BEPU_{b,t-1}$ is defined in equation (2). All specifications include quarter and bank fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 10: Heterogeneity by bank and firm characteristics

	Bank characteristics			**Firm characteristics**	
Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1) High NPL	(2) Low excess liquidity	(3) Low capital	(4) High PD	(5) Net exporter to US
$BEPU_{b,t-1} \times$ Char	−0.00941*	−0.0102*	−0.00742	−0.00415**	−0.00321***
	(0.00528)	(0.00424)	(0.00536)	(0.00169)	(0.00116)
$BEPU_{b,t-1}$	−0.00215	0.00193	−0.00181	−0.00484**	−0.00536**
	(0.00284)	(0.00419)	(0.00295)	(0.00240)	(0.00320)
Observations	546,993	546,993	546,993	470,901	581,292
R^2	0.783	0.783	0.783	0.764	0.745
Firm × Quarter FE	Yes	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimates of equation (9). The dependent variable is the logarithm of new total credit. Columns (1)–(3) interact $BEPU_{b,t-1}$ with indicators for banks with NPL ratios above the 75th percentile, excess liquidity over total assets below the 25th percentile, and CET1 ratios below the 25th percentile of the cross-sectional distribution. Columns (4)–(5) interact $BEPU_{b,t-1}$ with indicators for firms with a probability of default above the 75th percentile, where PD is the exposure-weighted average of the internal risk assessments of the firm's lenders, and for firms in country–sector (NACE two-digit) cells whose net exports to the United States, defined in Section 3 using OECD ICIO data, lie above the 75th percentile. Lower-order terms are absorbed by the fixed effects or included. All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 11: Asset-side versus liability-side exposure to U.S. policy uncertainty

Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)
$BEPU^{\text{Assets}}_{b,t-1}$	-0.00617^{***}	-0.00734^{***}	-0.00597^{***}
	(0.00221)	(0.00208)	(0.00209)
US Liabilities/Total Liabilities$_{b,t-1}$		0.0148^{*}	
		(0.00882)	
$BEPU^{\text{Liabilities}}_{b,t-1}$			-0.00128
			(0.00955)
Observations	599,637	599,637	599,637
R^2	0.774	0.774	0.774
Firm × Quarter FE	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes

Notes: The table compares the role of banks' U.S. asset and U.S. liability exposure in the transmission of U.S. policy uncertainty to lending. The dependent variable is the logarithm of new total credit. $BEPU^{\text{Assets}}_{b,t-1}$ is the baseline measure of equation (2). $BEPU^{\text{Liabilities}}_{b,t-1}$ is constructed analogously using the share of U.S. liabilities in total liabilities (FINREP F20.06), where total liabilities include deposits, derivatives, and short positions. Column (2) adds the lagged U.S. liability share as a control; column (3) includes both measures jointly. All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 12: Direct asset exposure versus borrowers' trade exposure to the United States

Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)
$BEPU^{\text{Assets}}_{b,t-1}$	-0.00617^{***}	-0.00861^{***}	-0.00851^{***}
	(0.00221)	(0.00217)	(0.00199)
$BEPU^{\text{US Exports}}_{b,t-1}$		-0.0636^{**}	
		(0.0261)	
$BEPU^{\text{US Net Exports}}_{b,t-1}$			-0.149^{***}
			(0.0571)
Observations	599,637	599,637	599,637
R^2	0.774	0.774	0.774
Firm × Quarter FE	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes

Notes: The table compares banks' direct U.S. asset exposure with their indirect exposure to the United States through the trade exposure of their borrowers, following Paravisini, Rappoport and Schnabl (2023). The dependent variable is the logarithm of new total credit. $BEPU^{\text{Assets}}_{b,t-1}$ is the baseline measure. $BEPU^{\text{US Exports}}_{b,t-1}$ weights each bank's lending across country–sector cells by the cells' share of exports going to the United States; $BEPU^{\text{US Net Exports}}_{b,t-1}$ uses net exports to the United States instead (Section 5.6). All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 13: Firm-level effects of lenders' exposure to U.S. policy uncertainty

	(1) Bank loans	(2) Total debt	(3) Investment	(4) Employment
$Exposed_{f,t-1}$	-0.0245^{***}	-0.000889^{***}	-0.00604^{***}	-0.00157^{***}
	(0.000420)	(0.0000501)	(0.000515)	(0.000515)
Observations	1,247,406	1,247,293	1,185,056	1,174,643
R^2	0.0179	0.0853	0.136	0.0367
Firm FE	Yes	Yes	Yes	Yes
Industry × Country × Year FE	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes

Notes: The table reports estimates of equation (12) at annual frequency over 2014–2023. The dependent variable is the annual log change in new bank loans (column 1), total debt (column 2), fixed assets (column 3), and employment (column 4). $Exposed_{f,t-1}$ is the average *BEPU* of the firm's lenders, weighted by each lender's share in the firm's outstanding bank borrowing in the previous year (equation (11)). All specifications include firm and industry (NACE two-digit)×country×year fixed effects and lagged firm controls (log total assets, cash over total assets, tangibility, interest coverage ratio, sales growth). Standard errors, in parentheses, are clustered at the firm level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 14: Firm-level effects: net exporters to the United States

	(1) Bank loans	(2) Total debt	(3) Investment	(4) Employment
$Exposed_{f,t-1} \times \text{Net Exporter}_f$	-0.00229^{**}	-0.000229^{**}	-0.00166^{*}	0.000516
	(0.00107)	(0.000102)	(0.00100)	(0.00127)
$Exposed_{f,t-1}$	-0.0234^{***}	-0.000814^{***}	-0.00502^{***}	-0.00161^{**}
	(0.000403)	(0.0000581)	(0.000548)	(0.000645)
Observations	1,213,322	1,213,216	1,152,886	1,149,000
R^2	0.0178	0.0862	0.137	0.0230
Firm FE	Yes	Yes	Yes	Yes
Industry × Country × Year FE	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes

Notes: The table reports estimates of equation (12) augmented with an interaction between $Exposed_{f,t-1}$ and Net Exporter_f, an indicator for firms in country–sector (NACE two-digit) cells with positive net exports to the United States, defined in Section 3 using OECD ICIO data. The dependent variables and controls are as in Table 13. Standard errors, in parentheses, are clustered at the firm level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 15: Bank lending and alternative sources of uncertainty and risk

	US–China trade tension		**US monetary policy uncertainty**		**Geopolitical risk**	
Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)	(4)	(5)	(6)
$BRisk_{b,t-1}$	−0.00397*	0.00243	−0.00450***	−0.00110	−0.00916***	−0.00566*
	(0.00245)	(0.00218)	(0.00255)	(0.00190)	(0.00279)	(0.00153)
$BEPU_{b,t-1}$		−0.00786***		−0.00669**		−0.00555**
		(0.00202)		(0.00270)		(0.00247)
Observations	587,058	587,058	554,840	554,840	587,058	587,058
R^2	0.775	0.775	0.774	0.774	0.774	0.774
Firm × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimates of equation (13). The dependent variable is the logarithm of new total credit. $BRisk_{b,t-1}$ is constructed as $BEPU_{b,t-1}$ in equation (2), replacing the U.S. EPU index with the U.S.–China trade tension index of Rogers, Sun and Sun (2024) (columns 1–2, available until 2023Q4), the U.S. monetary policy uncertainty index of Husted, Rogers and Sun (2020) (columns 3–4, available until 2024Q2), and the geopolitical risk index of Caldara and Iacoviello (2022) (columns 5–6). All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 16: Loan rates and alternative sources of uncertainty and risk

	US–China trade tension		**US monetary policy uncertainty**		**Geopolitical risk**	
Dependent variable: Loan rate$_{b,f,t}$	(1)	(2)	(3)	(4)	(5)	(6)
$BRisk_{b,t-1}$	0.00140***	0.000935**	0.00127***	0.000928***	0.00111***	0.000431
	(0.000236)	(0.000380)	(0.000205)	(0.000243)	(0.000320)	(0.000300)
$BEPU_{b,t-1}$		0.000564*		0.00674***		0.00108***
		(0.000333)		(0.00239)		(0.00229)
Observations	583,084	583,084	540,157	540,157	570,705	570,705
R^2	0.613	0.613	0.613	0.613	0.614	0.615
Firm × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports estimates of equation (13) with the annualized loan rate as dependent variable. $BRisk_{b,t-1}$ is defined in the notes to Table 15. All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 17: Global policy uncertainty and the global financial cycle

	Total credit (log)			**Loan rate**		
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{BGEPU}_{b,t-1}$	-0.0102^{***}		-0.00392^{*}	0.00223^{***}		0.00203^{***}
	(0.00249)		(0.00230)	(0.000322)		(0.000437)
$\text{BVIX}_{b,t-1}$		-0.0499^{***}	-0.0269		0.00731^{***}	0.00132
		(0.0164)	(0.0270)		(0.00149)	(0.00169)
Observations	599,637	599,637	599,637	583,084	583,084	583,084
R^2	0.770	0.774	0.774	0.613	0.612	0.613
Firm × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table examines whether the effect of global policy uncertainty on bank lending is distinct from the global financial cycle. $\text{BGEPU}_{b,t-1}$ interacts banks' average exposure to countries outside the euro area over the previous four quarters (in percent of total assets) with lagged global EPU, the GDP-PPP-weighted average of the EPU indices of fourteen non-euro-area countries. $\text{BVIX}_{b,t-1}$ interacts the same extra-euro-area exposure with the lagged VIX. The dependent variable is the logarithm of new total credit in columns (1)–(3) and the annualized loan rate in columns (4)–(6). All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Appendix

A.1 Additional figures

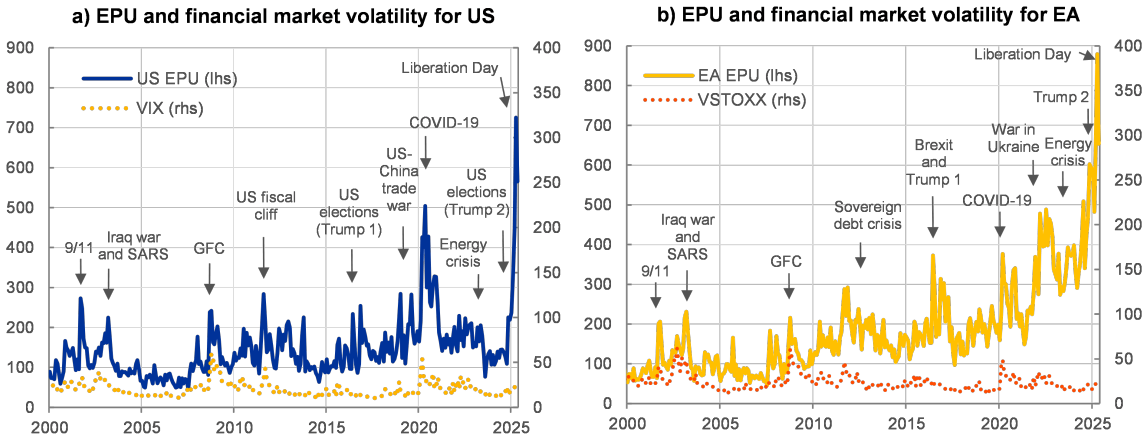


Figure A1: Economic policy uncertainty in the United States and the euro area

Notes: The left panel plots the U.S. EPU index (left axis) and the VIX (right axis); the right panel plots the euro area EPU index (left axis) and the VSTOXX (right axis), monthly, 2000–2025. EPU indices are from Baker, Bloom and Davis (2016).

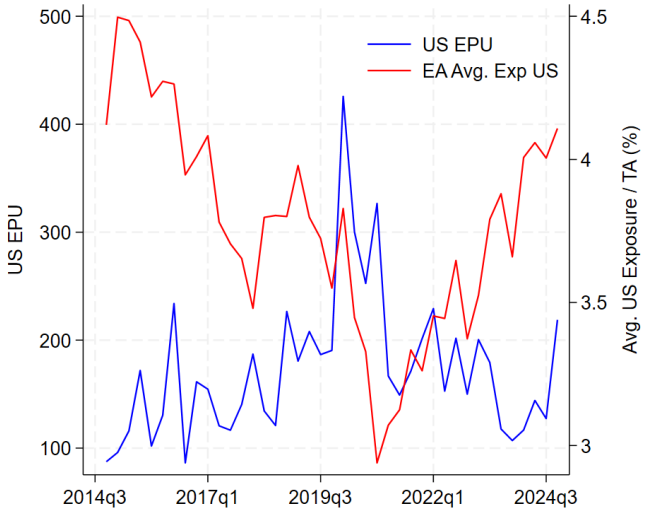


Figure A2: U.S. policy uncertainty and the average U.S. exposure of euro area banks

Notes: The figure plots the quarterly average of the U.S. EPU index (left axis) and the average U.S. exposure of the 79 euro area banking groups in the sample, in percent of total assets (right axis), from 2014Q4 to 2024Q4. The correlation between the two series is −0.37. Sources: Baker, Bloom and Davis (2016); FINREP F20.04.

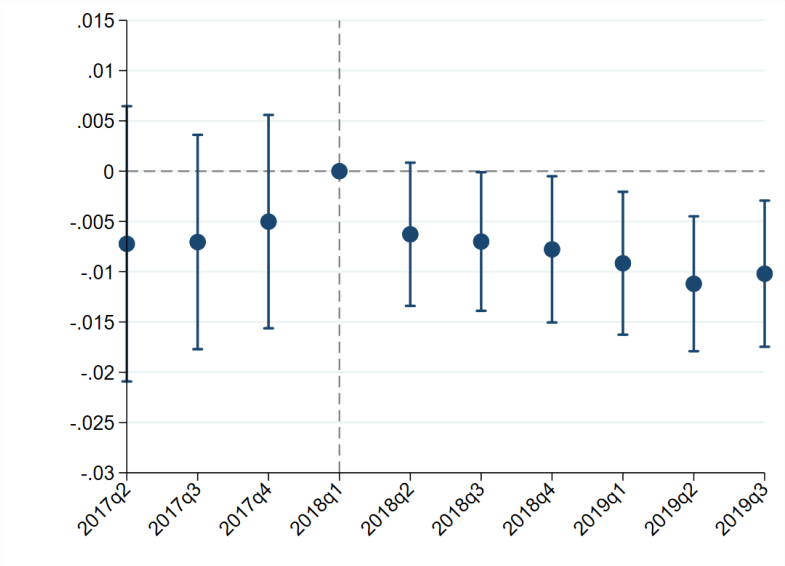


Figure A3: Dynamic effects of the 2018 U.S.–China trade war on lending to euro area firms

Notes: The figure reports event-study coefficients, with 90 percent confidence intervals, from the specification in the notes to Figure 4 applied to the escalation of the U.S.–China trade war. The omitted quarter is 2018Q1; U.S. exposure is measured in 2017Q1. The dependent variable is the logarithm of total new credit. The specification includes firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors are clustered at the bank level.

A.2 Other uncertainty episodes

Table A1: 2016 U.S. election: effects on bank lending and credit conditions in the euro area

	Bank lending			Credit conditions		
	(1) Total credit	(2) Term loans	(3) Credit lines	(4) Exit	(5) Loan maturity	(6) Loan rate
Post × US Exp_b	−0.0192**	−0.0205**	−0.0185	0.0069	−0.011***	0.00035***
	(0.00977)	(0.00975)	(0.07667)	(0.005)	(0.003)	(0.00012)
Observations	318,551	318,551	6,609	318,551	201,240	211,172
R^2	0.755	0.771	0.908	0.761	0.982	0.979
Firm × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports difference-in-differences estimates of equation (3) around the 2016 U.S. election. The pre-election period is 2015Q4–2016Q3 and the post-election period is 2016Q4–2017Q4. US Exp_b is U.S. assets over total assets as of 2015Q3. Dependent variables, controls, and fixed effects are as in Table 3. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A2: 2016 U.S. election: bank-level effects

	(1) **Total**	(2) **US**	(3) **EA**	(4) **RoW**
Panel A: Outstanding loans to non-financial corporations / total assets (log)				
Post × US Exp_b	−0.0643***	−0.0114	−0.0391***	−0.0151
	(0.0139)	(0.0150)	(0.0108)	(0.0180)
Observations	541	414	539	531
Adj. R^2	0.923	0.978	0.944	0.943
Panel B: Central government securities / total assets (log)				
Post × US Exp_b	0.00820	−0.00268	0.0104*	−0.000747
	(0.00599)	(0.0128)	(0.00601)	(0.00732)
Observations	475	386	475	451
Adj. R^2	0.969	0.916	0.973	0.970
Panel C: Borrower probability of default (log)				
Post × US Exp_b		0.00987	0.0135***	0.00809
		(0.0113)	(0.00424)	(0.00594)
Observations		342	379	379
Adj. R^2		0.841	0.969	0.956
Quarter FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes

Notes: The table reports bank-level difference-in-differences estimates of equation (6) around the 2016 U.S. election, for the whole balance sheet (column 1) and by counterparty region (columns 2–4). In Panel C, PDs are weighted by the bank's credit exposure to each country and the sample period is 2015–2018. US Exp_b is U.S. assets over total assets before the election. All specifications include quarter and bank fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A3: 2020 U.S. election: effects on bank lending and credit conditions in the euro area

	Bank lending			Credit conditions		
	(1) Total credit	(2) Term loans	(3) Credit lines	(4) Exit	(5) Loan maturity	(6) Loan rate
Post × US Exp_b	0.154***	0.228***	−0.344	0.00410	0.252***	−0.000526**
	(0.0265)	(0.0390)	(0.546)	(0.0108)	(0.0278)	(0.000262)
Observations	19,203	19,160	1,155	19,203	18,266	16,221
R^2	0.657	0.641	0.814	0.543	0.702	0.639
Firm × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports difference-in-differences estimates of equation (3) around the 2020 U.S. election. The pre-election period is 2019Q4–2020Q3 and the post-election period is 2020Q4–2021Q4. US Exp_b is U.S. assets over total assets as of 2019Q3. Dependent variables, controls, and fixed effects are as in Table 3. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A4: 2018 U.S.–China trade war: effects on credit and loan terms

	(1) Total credit	(2) Term loans	(3) Loan rate
Post × US Exp_b	−0.00664***	−0.006495***	0.000190*
	(0.00290)	(0.00303)	(0.000102)
Observations	2,490,378	2,490,378	1,925,106
R^2	0.942	0.843	0.931
Firm × Quarter FE	Yes	Yes	Yes
Bank × Firm FE	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes

Notes: The table reports difference-in-differences estimates of equation (3) around the escalation of the U.S.–China trade war. The pre-period is 2017Q2–2018Q1 and the post-period is 2018Q2–2019Q4. US Exp_b is U.S. assets over total assets as of 2017Q1. Regressions are at the loan level; the loan-rate regression additionally controls for secured-loan and floating-rate indicators. All specifications include firm×quarter and bank×firm fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A5: 2024 U.S. election: borrower probability of default by region, continuous exposure

Dependent variable: $\log(\text{PD})_{b,c,t}$	(1) US	(2) EA	(3) RoW
Post × US Exp_b	0.0325**	0.00930*	0.00727
	(0.0130)	(0.00534)	(0.00489)
Observations	263	289	289
Adj. R^2	0.892	0.937	0.966
Quarter FE	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes

Notes: The table reports bank-level difference-in-differences estimates around the 2024 U.S. election in which the dependent variable is the logarithm of the exposure-weighted average PD of the bank's borrowers in each region, and the treatment is the continuous U.S. exposure measure US Exp_b (U.S. assets over total assets as of 2023Q3). Moving from the 25th to the 75th percentile of U.S. exposure implies an increase in the PD of U.S. (euro area) borrowers of about 21 (6) percent. All specifications include quarter and bank fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

A.3 Properties of *BEPU* and additional bank-level results

Table A6: Variance decomposition of *BEPU*

	Baseline	With bank controls
Share of variance common across banks within a quarter	0.107	0.269
Share of variance across banks within a quarter	0.893	0.731
Variance share explained by bank and quarter FE	0.748	0.785
Residual variance share (bank and quarter FE)	0.252	0.215

Notes: Let $\bar{b}_t$ denote the cross-sectional mean of $BEPU_{b,t}$ in quarter t and $\tilde{b}_{b,t} = BEPU_{b,t} - \bar{b}_t$ the within-quarter deviation. The first row reports $\mathrm{Var}(\bar{b}_t)/\mathrm{Var}(BEPU_{b,t})$ and the second row $\mathrm{Var}(\tilde{b}_{b,t})/\mathrm{Var}(BEPU_{b,t})$. The last two rows report the explained and residual variance shares from a regression of $BEPU_{b,t}$ on bank and quarter fixed effects. The second column first residualizes $BEPU_{b,t}$ on the bank controls.

Table A7: *BEPU* and bank characteristics

Dependent variable: $BEPU_{b,t}$	CET1 ratio	ROA	LCR	Total assets (log)	G-SII
Coefficient	18.50^{***}	-1.709^{***}	-0.555^{***}	0.918^{***}	3.827^{***}
	(1.641)	(0.355)	(0.126)	(0.122)	(0.394)
Observations			2,268		
R^2			0.226		
Bank FE			Yes		
Quarter FE			Yes		

Notes: The table reports the coefficients from a single bank–quarter regression of $BEPU_{b,t}$ on the CET1 ratio, return on assets, the liquidity coverage ratio, the logarithm of total assets, and an indicator for global systemically important institutions (G-SII), with bank and quarter fixed effects. Standard errors are in parentheses. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A8: *BEPU* and new lending: bank-level regressions

	Total credit (log)		**Term loans (log)**	
	(1)	(2)	(3)	(4)
$BEPU_{b,t}$	0.00260	0.000741	0.00263	0.00122
	(0.00423)	(0.00441)	(0.00424)	(0.00459)
$BEPU_{b,t-1}$	-0.00584^{*}	-0.00615^{*}	-0.00706^{*}	-0.00732^{*}
	(0.00345)	(0.00367)	(0.00402)	(0.00417)
Observations	2,158	2,158	2,158	2,158
R^2	0.951	0.952	0.949	0.950
Quarter FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Bank controls		Yes		Yes

Notes: The table reports bank–quarter regressions of the logarithm of new total credit and new term loans to euro area firms on contemporaneous and lagged *BEPU*, with bank and quarter fixed effects. Standard errors, in parentheses, are clustered at the bank level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A9: *BEPU* and borrower credit risk by region

Dependent variable: $\log(\text{PD})_{b,c,t}$	(1) US	(2) EA	(3) RoW
$BEPU_{b,t-1}$	0.0160**	0.00658*	0.0106**
	(0.00812)	(0.00378)	(0.00428)
Observations	1,445	1,522	1,520
Adj. R^2	0.851	0.860	0.856
Quarter FE	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
Bank controls	Yes	Yes	Yes

Notes: The table reports estimates of equation (8) in which the dependent variable is the logarithm of the exposure-weighted average PD of the bank's borrowers in the United States, the euro area, and the rest of the world. All specifications include quarter and bank fixed effects and lagged bank controls. Standard errors, in parentheses, are clustered at the bank level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A10: *BEPU* and the formation of new lending relationships

Dependent variable: $Entry_{b,f,t}$	(1)	(2)	(3)
$BEPU_{b,t-1}$	−0.0000838	0.000518	−0.000135
	(0.000373)	(0.000382)	(0.000389)
Observations	1,841,988	1,329,671	3,547,826
R^2	0.429	0.412	0.352
Bank FE	Yes		Yes
Firm FE			Yes
Firm × Quarter FE	Yes	Yes	
Bank × Firm FE		Yes	
Industry × Location × Quarter FE			Yes
Bank controls	Yes	Yes	Yes

Notes: Entry is an indicator equal to one if a bank–firm relationship did not exist in quarter $t-1$ and is established in quarter t. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A11: Bank characteristics: alternative fixed-effect structures

Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)	(4)	(5)	(6)
$BEPU_{b,t-1}$	−0.00472	0.00152	−0.00486	−0.00331	0.00388	−0.00349
	(0.00369)	(0.00564)	(0.00379)	(0.00385)	(0.00588)	(0.00398)
$BEPU_{b,t-1}$ × High NPL	−0.0133**			−0.0149**		
	(0.00639)			(0.00660)		
$BEPU_{b,t-1}$ × Low excess liquidity		−0.0140**			−0.0160***	
		(0.00573)			(0.00592)	
$BEPU_{b,t-1}$ × Low capital			−0.00835			−0.00961
			(0.00646)			(0.00669)
Observations	592,579	613,036	613,036	934,964	985,966	985,966
R^2	0.634	0.635	0.634	0.710	0.709	0.709
Firm × Quarter FE	Yes	Yes	Yes			
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE				Yes	Yes	Yes
Industry × Location × Quarter FE				Yes	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table re-estimates columns (1)–(3) of Table 10 with alternative fixed-effect structures, as indicated in the bottom rows: bank and firm×quarter fixed effects in columns (1)–(3), and firm, bank, and industry×location×quarter fixed effects in columns (4)–(6). Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A12: Liability-weighted *BEPU* and Asset-weighted *BEPU*

Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)
$BEPU^{\text{Liabilities}}_{b,t-1}$	-0.00810^{***}	-0.00478^{***}	-0.00493^{***}
	(0.00188)	(0.00182)	(0.00181)
Observations	671,477	599,637	994,884
R^2	0.626	0.774	0.791
Bank FE	Yes		Yes
Firm FE			Yes
Firm × Quarter FE	Yes	Yes	
Bank × Firm FE		Yes	
Industry × Location × Quarter FE			Yes
Bank controls	Yes	Yes	Yes

Notes: $BEPU^{\text{Liabilities}}_{b,t-1}$ is constructed as in equation (2) using the share of U.S. deposits from credit institutions, other financial institutions, non-financial corporations, and households in total deposits (FINREP F20.06) in place of the U.S. asset share. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A13: Trade-weighted *BEPU* and Asset-weighted *BEPU*

Dependent variable: $\log(\text{Term loans})_{b,f,t}$	(1)	(2)	(3)	(4)	(5)	(6)
$BEPU^{\text{US Exports}}_{b,t-1}$	−0.0433	-0.0488^{*}	-0.0588^{*}			
	(0.0330)	(0.0280)	(0.0318)			
$BEPU^{\text{US Net Exports}}_{b,t-1}$				-0.119^{*}	-0.123^{*}	-0.146^{*}
				(0.0708)	(0.0650)	(0.0751)
Observations	665,789	595,111	983,995	665,789	595,111	983,995
R^2	0.610	0.767	0.790	0.610	0.768	0.790
Bank FE	Yes		Yes	Yes		Yes
Firm FE			Yes			Yes
Firm × Quarter FE	Yes	Yes		Yes	Yes	
Bank × Firm FE		Yes			Yes	
Industry × Location × Quarter FE			Yes			Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of new term loans. $BEPU^{\text{US Exports}}_{b,t-1}$ and $BEPU^{\text{US Net Exports}}_{b,t-1}$ are defined in Section 5.6. A one-standard-deviation increase in $BEPU^{\text{US Exports}}$ (2.16) reduces new term loans by about 10.5 percent; a one-standard-deviation increase in $BEPU^{\text{US Net Exports}}$ (0.99) reduces them by about 12 percent. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

A.4 Robustness of the baseline panel estimates

Table A14: Robustness to the sample period and to the measurement of U.S. EPU

	Total credit			Term loans		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Excluding the COVID-19 period (2019Q4–2021Q1)						
$BEPU_{b,t-1}$	−0.00958***	−0.00519***	−0.00924***	−0.00895***	−0.00481***	−0.00832***
	(0.00164)	(0.00142)	(0.00150)	(0.00161)	(0.00141)	(0.00147)
Observations	543,930	487,246	856,572	539,682	483,926	846,639
R^2	0.639	0.793	0.719	0.629	0.791	0.717
Panel B: Residualized U.S. EPU						
$\widehat{BEPU}_{b,t-1}$	−0.00796***	−0.00573***	−0.00672***	−0.00943***	−0.00696***	−0.00843***
	(0.00211)	(0.00200)	(0.00221)	(0.00285)	(0.00264)	(0.00307)
Observations	671,477	599,637	1,070,536	666,713	595,831	1,059,352
R^2	0.626	0.774	0.702	0.611	0.768	0.694
Bank FE	Yes		Yes	Yes		Yes
Firm FE			Yes			Yes
Firm × Quarter FE	Yes	Yes		Yes	Yes	
Bank × Firm FE		Yes			Yes	
Industry × Location × Quarter FE			Yes			Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Panel A excludes 2019Q4–2021Q1 from the sample. Panel B replaces the U.S. EPU index with the residual from a regression of the index on real U.S. GDP growth, S&P 500 returns, the federal funds rate, and the VIX before constructing *BEPU*. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A15: Robustness to domestic economic policy uncertainty

Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)
$BEPU_{b,t-1}$	−0.00986***	−0.00597***	−0.00846***
	(0.00229)	(0.00204)	(0.00242)
Domestic $\text{EPU}_{b,t-1}$	0.000183*	−0.000235**	0.000216**
	(0.0000982)	(0.000101)	(0.0000941)
Observations	628,946	564,538	995,754
R^2	0.628	0.776	0.706
Bank FE	Yes		Yes
Firm FE			Yes
Firm × Quarter FE	Yes	Yes	
Bank × Firm FE		Yes	
Industry × Location × Quarter FE			Yes
Bank controls	Yes	Yes	Yes

Notes: Domestic EPU is the country EPU index of Baker, Bloom and Davis (2016) assigned according to the country of each bank's headquarters, available for banks headquartered in Germany, France, Greece, Ireland, Italy, and Spain. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A16: Robustness to the timing of U.S. EPU

Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)
Panel A: Lagged quarterly-average EPU (baseline)			
$BEPU^{\text{avg}}_{b,t-1}$	-0.0110^{***}	-0.00620^{***}	-0.0101^{***}
	(0.00174)	(0.00155)	(0.00165)
Panel B: Contemporaneous quarterly-average EPU			
$BEPU^{\text{avg}}_{b,t}$	-0.00969^{***}	-0.00647^{***}	-0.00844^{***}
	(0.00210)	(0.00194)	(0.00221)
Panel C: Lagged end-of-quarter EPU			
$BEPU^{\text{end}}_{b,t-1}$	-0.00919^{***}	-0.00589^{***}	-0.00801^{***}
	(0.00203)	(0.00188)	(0.00213)
Observations	671,477	599,637	1,070,536
R^2	0.627	0.774	0.702
Bank FE	Yes		Yes
Firm FE			Yes
Firm × Quarter FE	Yes	Yes	
Bank × Firm FE		Yes	
Industry × Location × Quarter FE			Yes
Bank controls	Yes	Yes	Yes

Notes: The three panels use, respectively, the lagged quarterly average, the contemporaneous quarterly average, and the lagged end-of-quarter value of the monthly U.S. EPU index in the construction of *BEPU*. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A17: Robustness to alternative asset-based measures of U.S. exposure

	Total loans and advances to US			**Loans and advances to US NFCs**		
Dependent variable: $\log(\text{Total credit})_{b,f,t}$	(1)	(2)	(3)	(4)	(5)	(6)
$BEPU_{b,t-1}$	-0.00906^{***}	-0.00593^{**}	-0.00767^{**}	-0.0279^{***}	-0.0213^{***}	-0.0255^{***}
	(0.00294)	(0.00260)	(0.00318)	(0.00511)	(0.00569)	(0.00555)
Observations	671,477	599,637	1,070,536	671,477	599,637	1,070,536
R^2	0.626	0.774	0.702	0.627	0.774	0.702
Bank FE	Yes		Yes	Yes		Yes
Firm FE			Yes			Yes
Firm × Quarter FE	Yes	Yes		Yes	Yes	
Bank × Firm FE		Yes			Yes	
Industry × Location × Quarter FE			Yes			Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: $BEPU_{b,t-1}$ is constructed as in equation (2) using, in place of total U.S. assets, total loans and advances to U.S. counterparties (columns 1–3) and loans and advances to U.S. non-financial corporations (columns 4–6), each over total assets. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A18: Robustness to additional bank balance-sheet controls

	(1) Total credit	(2) Term loans	(3) Total credit	(4) Term loans
$BEPU_{b,t-1}$	−0.00825***	−0.00978***	−0.00790***	−0.00930***
	(0.00223)	(0.00305)	(0.00229)	(0.00305)
CET1 ratio$_{b,t-1}$	−2.559	−2.633	−2.122	−2.436
	(1.806)	(2.288)	(1.758)	(1.878)
ROA$_{b,t-1}$	−0.0109	0.00446	−0.00113	0.0153
	(0.0641)	(0.0875)	(0.0655)	(0.0875)
LCR$_{b,t-1}$	0.00211	−0.0277	0.00315	
	(0.0410)	(0.0506)	(0.0401)	
Log total assets$_{b,t-1}$	0.525***	0.586***	0.520***	0.583***
	(0.0918)	(0.104)	(0.0946)	(0.0985)
NPL ratio$_{b,t-1}$			−0.00190	−0.00858
			(0.0121)	(0.0147)
Cash/TA$_{b,t-1}$				−0.00518
				(0.00320)
Observations	922,592	914,154	1,070,536	1,059,352
R^2	0.689	0.680	0.702	0.694
Firm FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Industry × Location × Quarter FE	Yes	Yes	Yes	Yes

Notes: The table reports the coefficients on the bank controls in the specification with firm, bank, and industry×location×quarter fixed effects, and adds the NPL ratio and cash over total assets to the control set. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A19: Robustness: loan-level regressions

	Total credit			**Term loans**		
	(1)	(2)	(3)	(4)	(5)	(6)
$BEPU_{b,t-1}$	−0.00519*	−0.00324*	−0.00450*	−0.00535*	−0.00332*	−0.00467*
	(0.00290)	(0.00193)	(0.00270)	(0.00290)	(0.00193)	(0.00270)
Observations	10,118,977	9,811,275	10,619,312	10,118,977	9,811,275	10,619,312
R^2	0.707	0.728	0.693	0.705	0.725	0.689
Bank FE	Yes		Yes	Yes		Yes
Firm FE			Yes			Yes
Firm × Quarter FE	Yes	Yes		Yes	Yes	
Bank × Firm FE		Yes			Yes	
Industry × Location × Quarter FE			Yes			Yes
Bank controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table re-estimates the baseline specification of Table 7 at the loan (instrument) level. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A20: Robustness: credit lines

Dependent variable: $\log(\text{Credit lines})_{b,f,t}$	(1)	(2)	(3)
$BEPU_{b,t-1}$	0.00189	0.0151	−0.00519
	(0.0138)	(0.0120)	(0.0134)
Observations	27,320	13,644	46,810
R^2	0.815	0.848	0.845
Bank FE	Yes		Yes
Firm FE			Yes
Firm × Quarter FE	Yes	Yes	
Bank × Firm FE		Yes	
Industry × Location × Quarter FE			Yes
Bank controls	Yes	Yes	Yes

Notes: The dependent variable is the logarithm of new credit lines at the bank–firm–quarter level. Fixed effects and controls are as in Table 7. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table A21: *BEPU* and term loans denominated in U.S. dollars

Dependent variable: $\log(\text{USD term loans})_{b,f,t}$	(1)	(2)
$BEPU_{b,t-1}$	−0.00417**	−0.00313*
	(0.00203)	(0.00177)
Observations	620,135	618,267
R^2	0.702	0.709
Bank FE	Yes	
Firm × Quarter FE	Yes	Yes
Bank × Firm FE		Yes
Bank controls	Yes	Yes
Loan controls	Yes	Yes

Notes: The dependent variable is the logarithm of new term loans denominated in U.S. dollars. The sample includes dollar loans to euro area and non-euro-area firms, extended through direct cross-border lending and through foreign branches. Loan controls are indicators for floating-rate and secured loans. Standard errors, in parentheses, are clustered at the bank–quarter level. *, **, *** denote significance at the 10, 5, and 1 percent levels.